



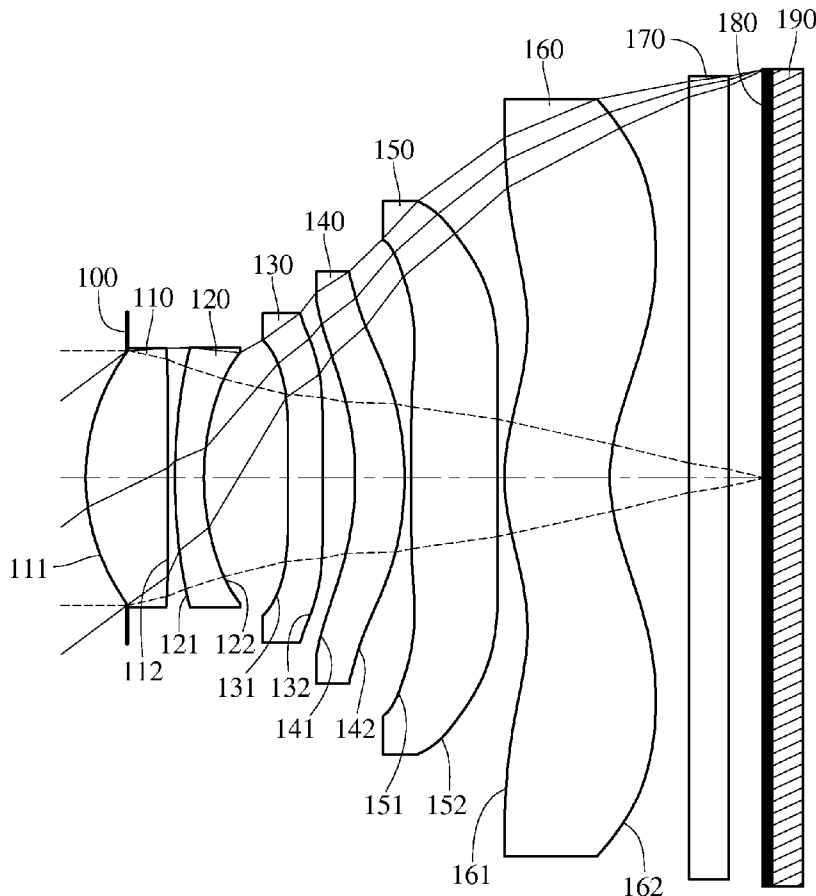
US 20160004039A1

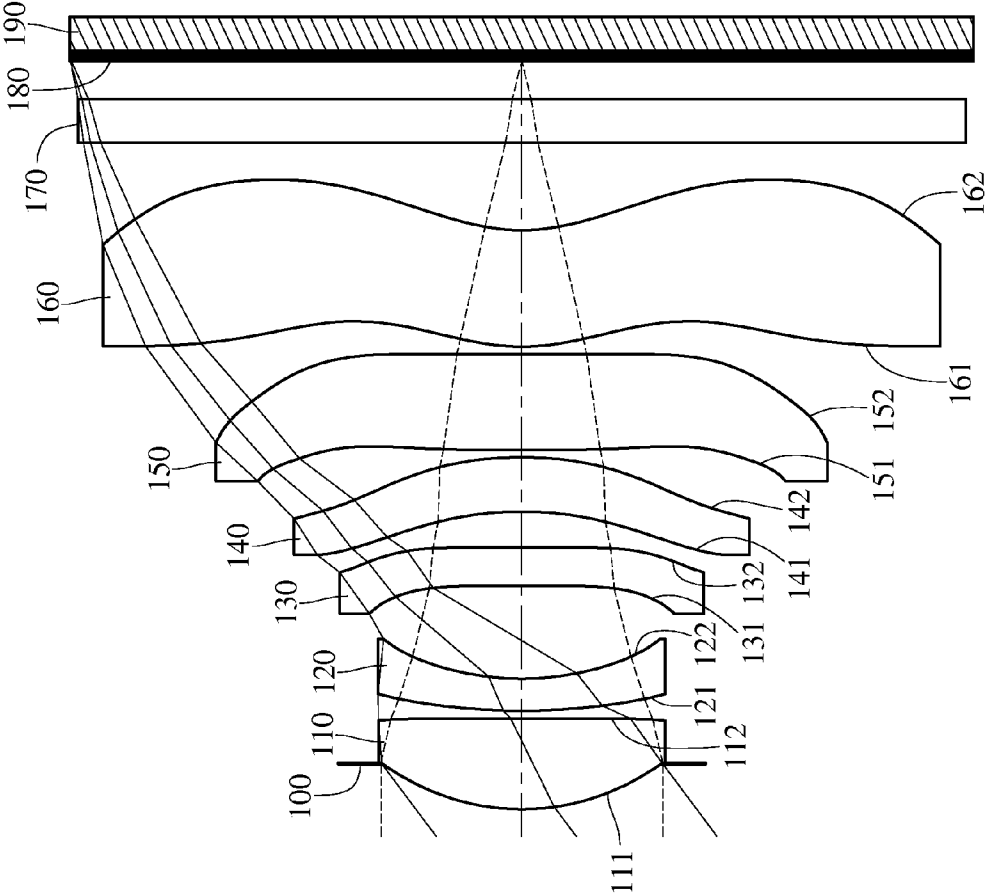
(19) **United States**(12) **Patent Application Publication**
CHEN(10) **Pub. No.: US 2016/0004039 A1**(43) **Pub. Date: Jan. 7, 2016**(54) **PHOTOGRAPHING OPTICAL LENS
ASSEMBLY, IMAGE CAPTURING UNIT AND
MOBILE TERMINAL**(52) **U.S. Cl.**
CPC **G02B 13/0045** (2013.01); **G02B 9/62**
(2013.01)(71) Applicant: **LARGAN Precision Co., Ltd.,**
Taichung (TW)(72) Inventor: **Wei-Yu CHEN**, Taichung (TW)(21) Appl. No.: **14/472,057**(22) Filed: **Aug. 28, 2014**(30) **Foreign Application Priority Data**

Jul. 4, 2014 (TW) 103123211

Publication Classification(51) **Int. Cl.**
G02B 13/00 (2006.01)
G02B 9/62 (2006.01)(57) **ABSTRACT**

A photographing optical lens system includes, in order from object side to image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element has negative refractive power. The fourth lens element has negative refractive power. The fifth lens element has refractive power. The sixth lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof.





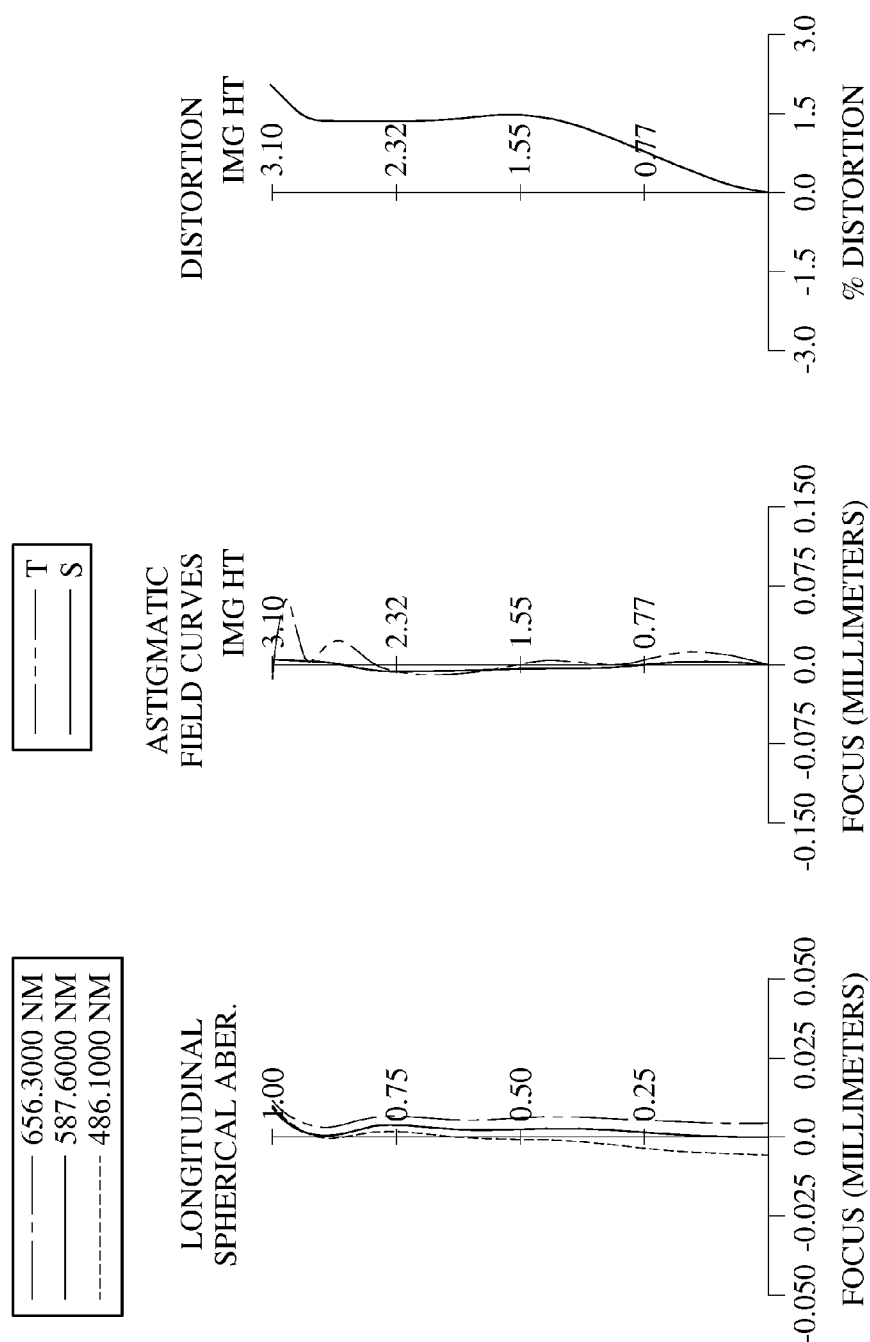
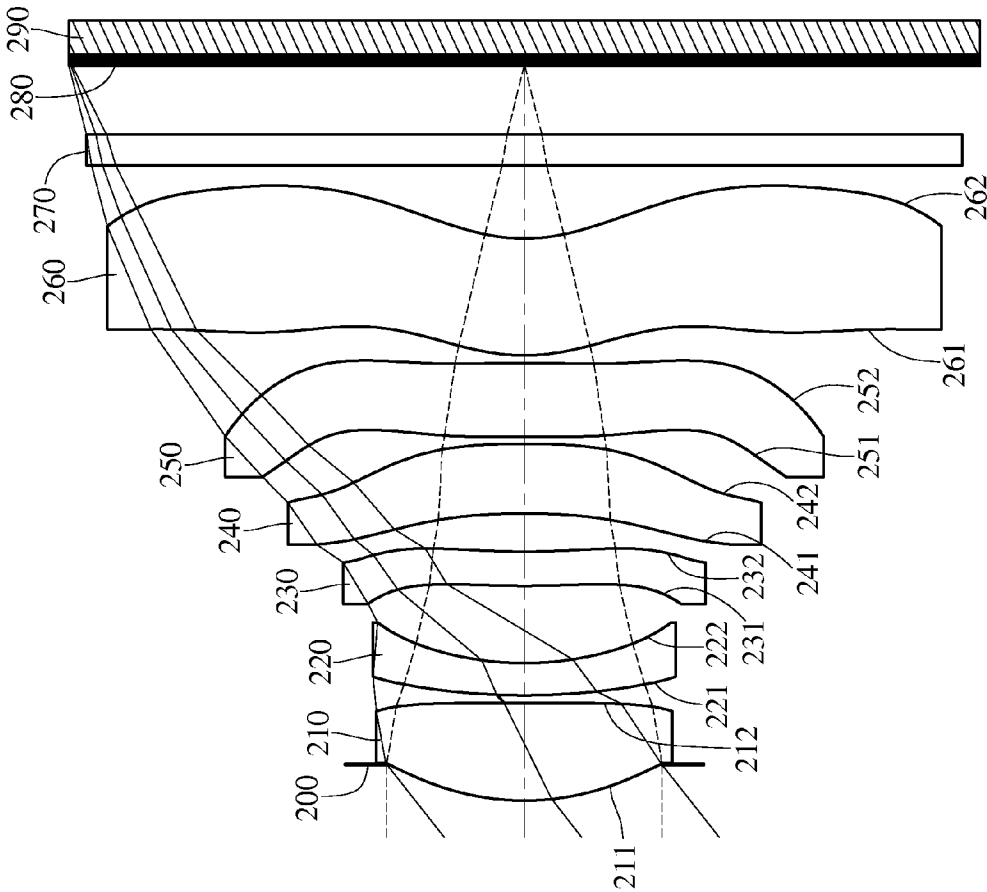


FIG. 2



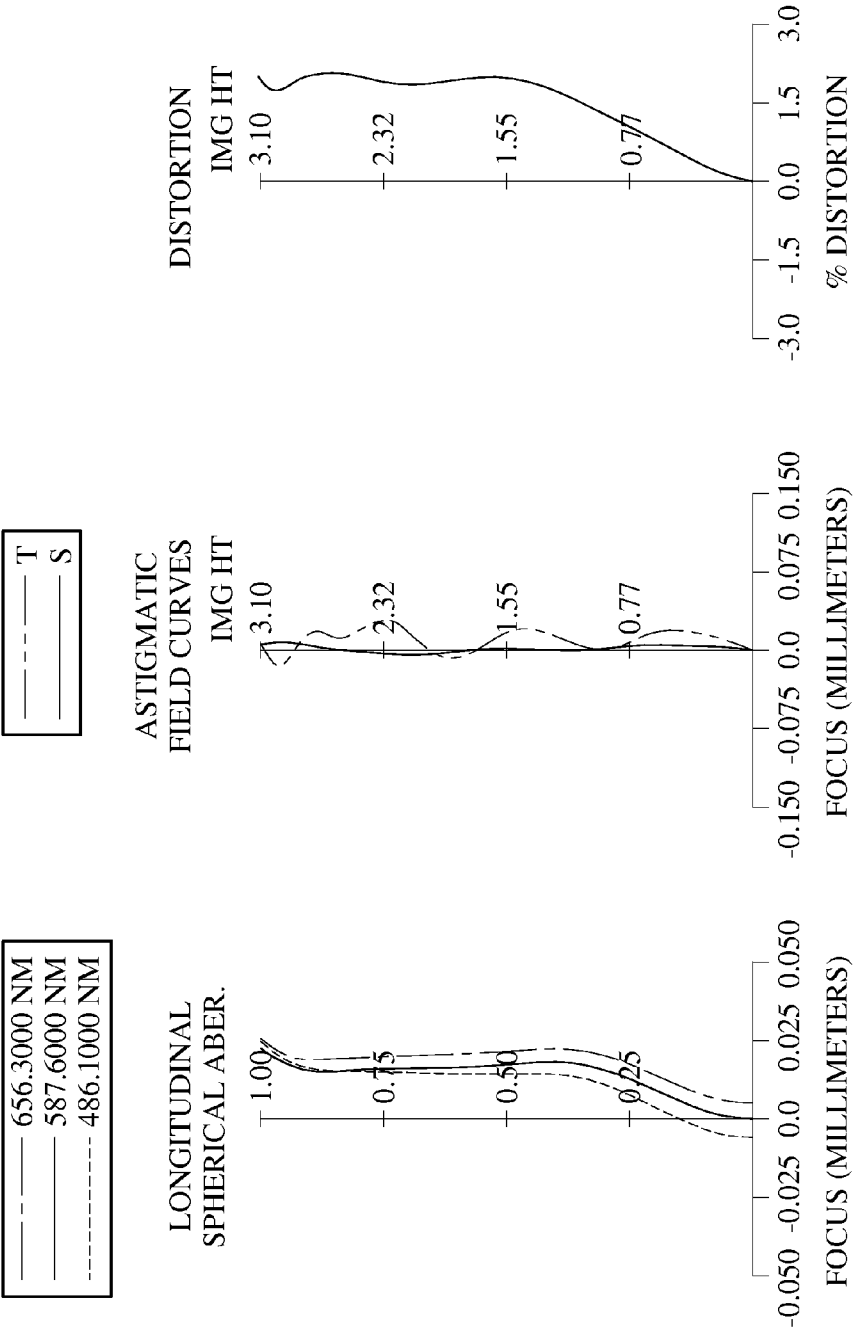


FIG. 4

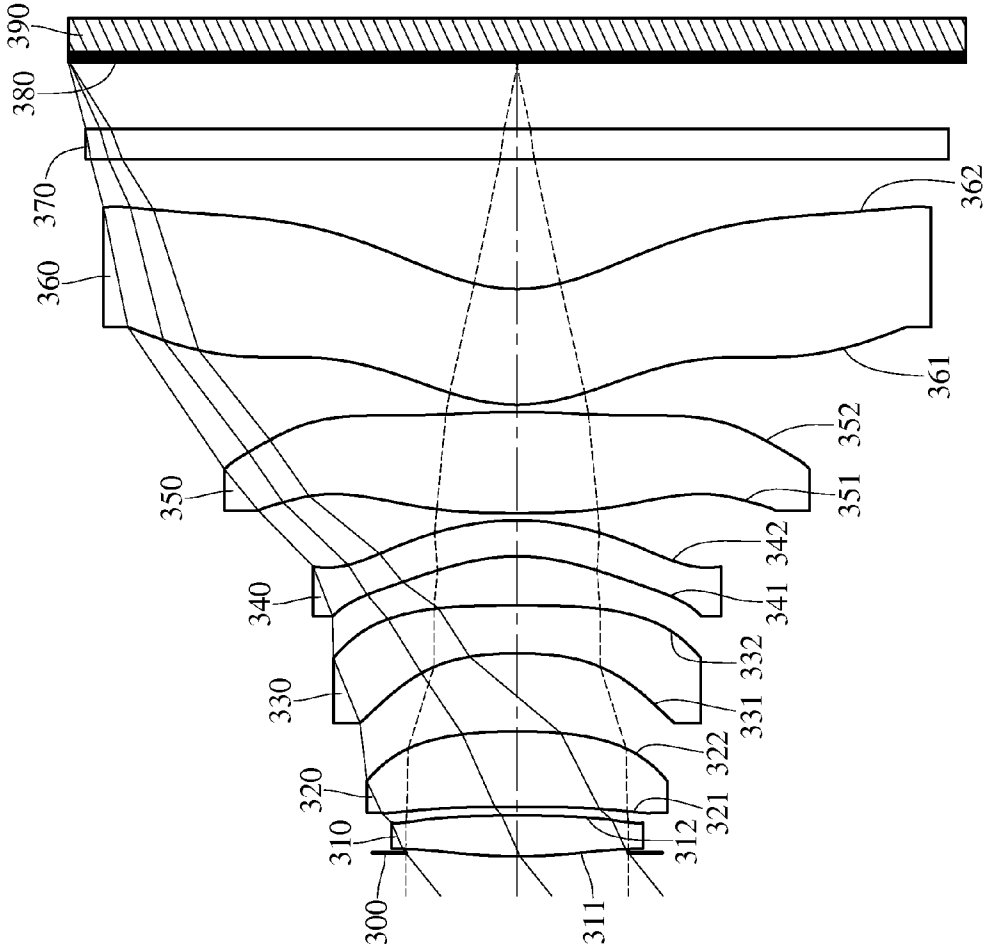
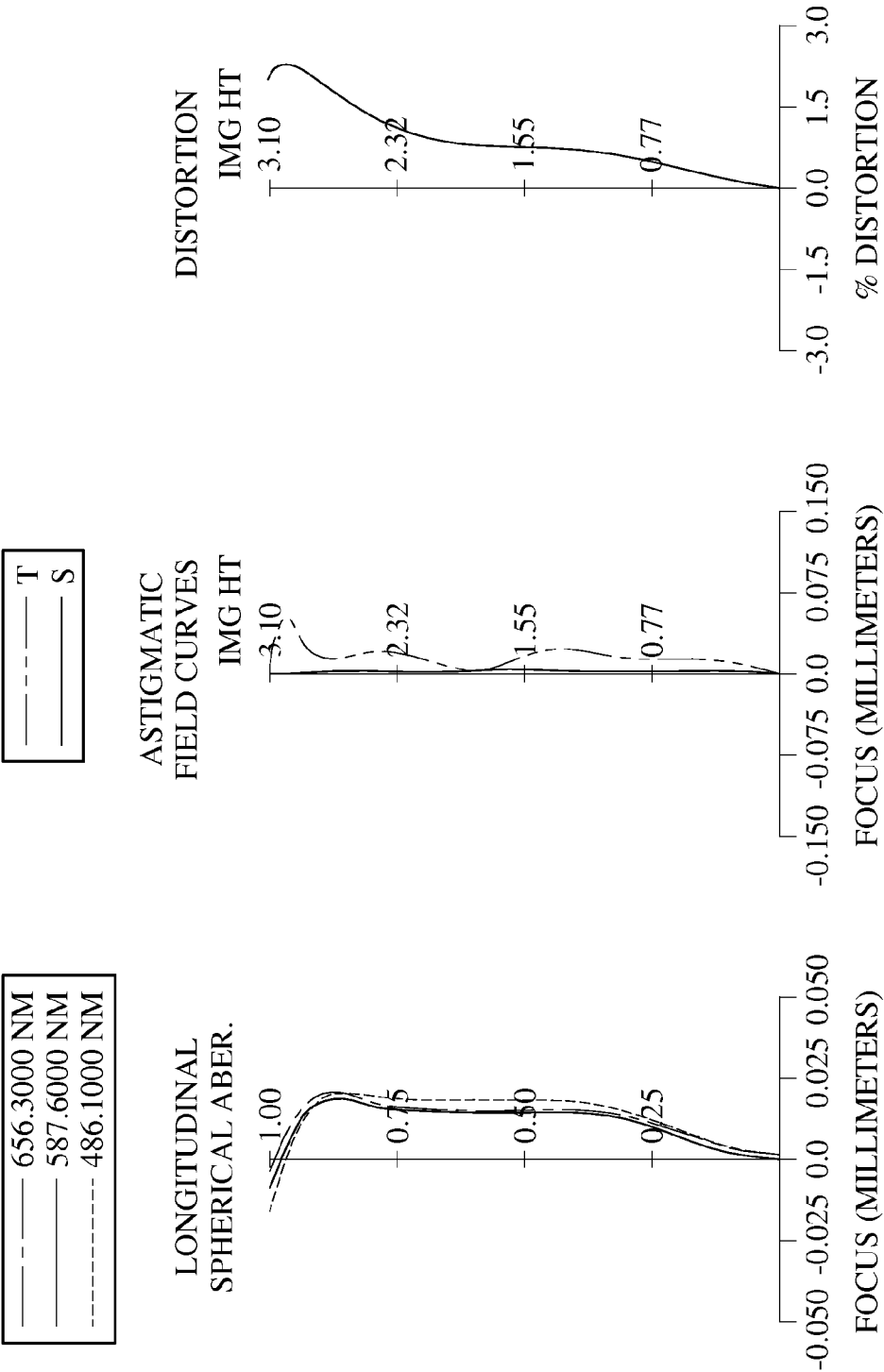


FIG. 5



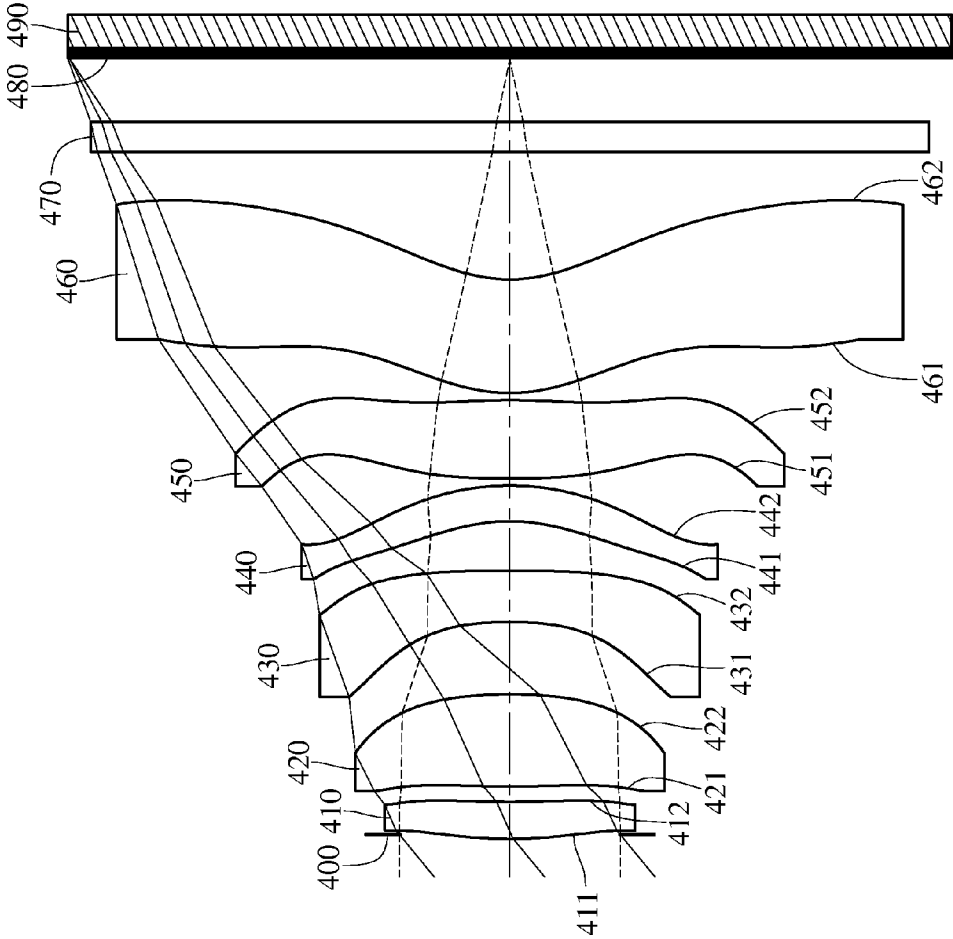
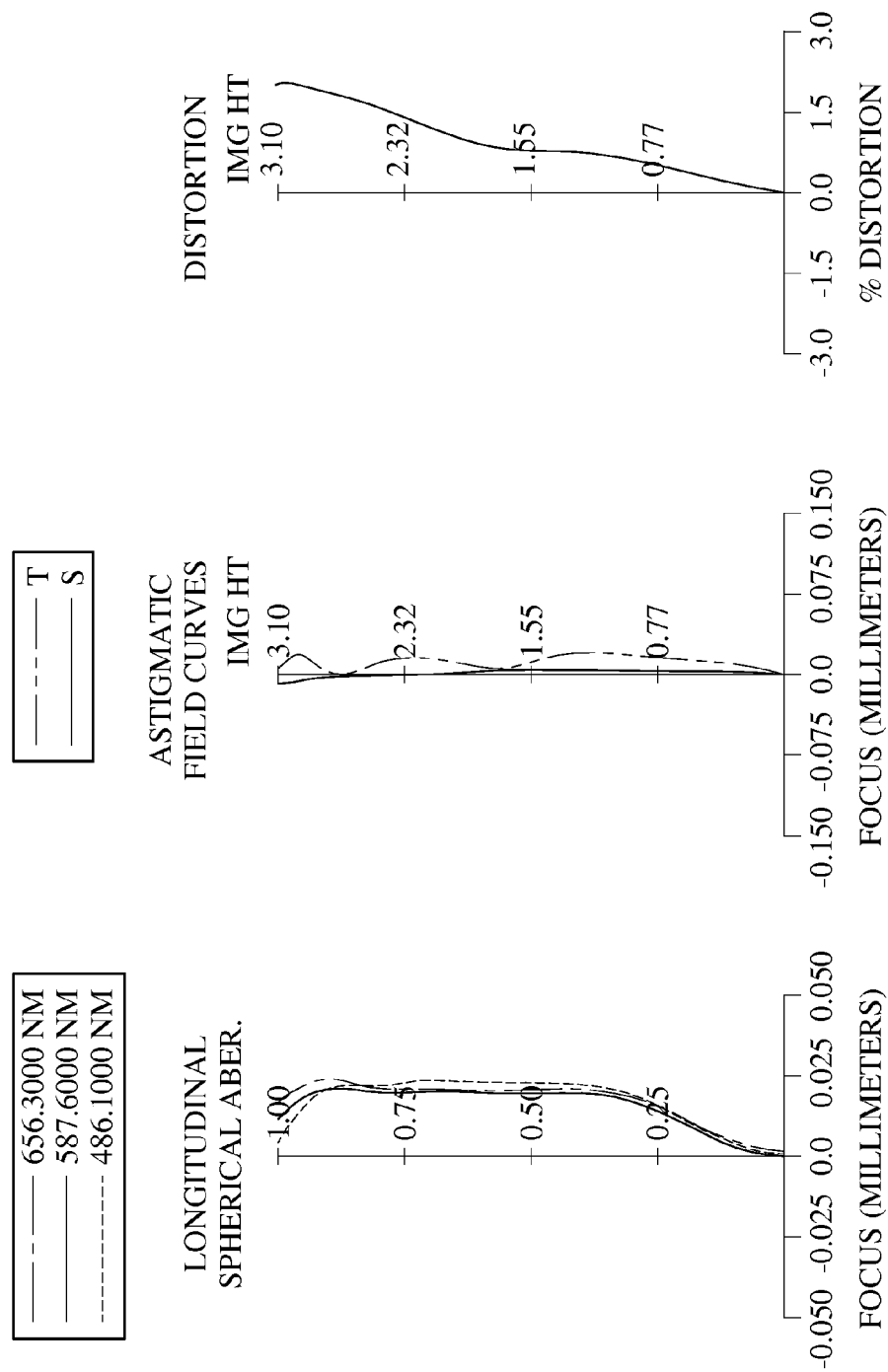


FIG. 7



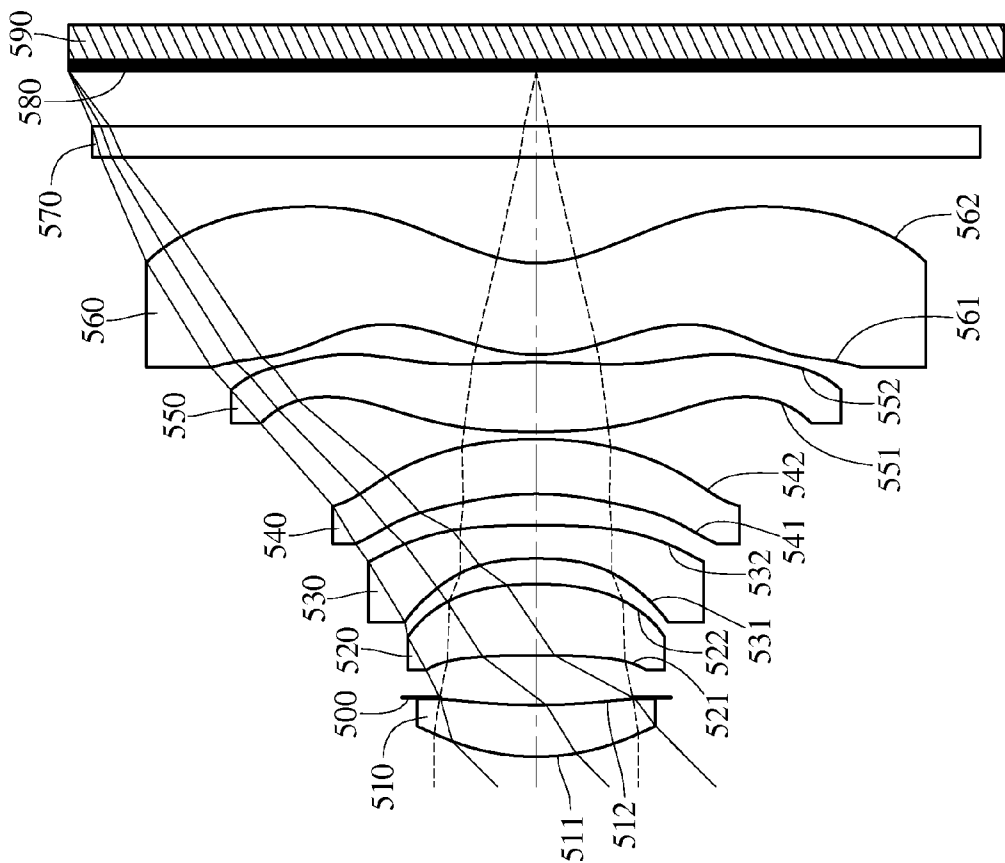


FIG. 9

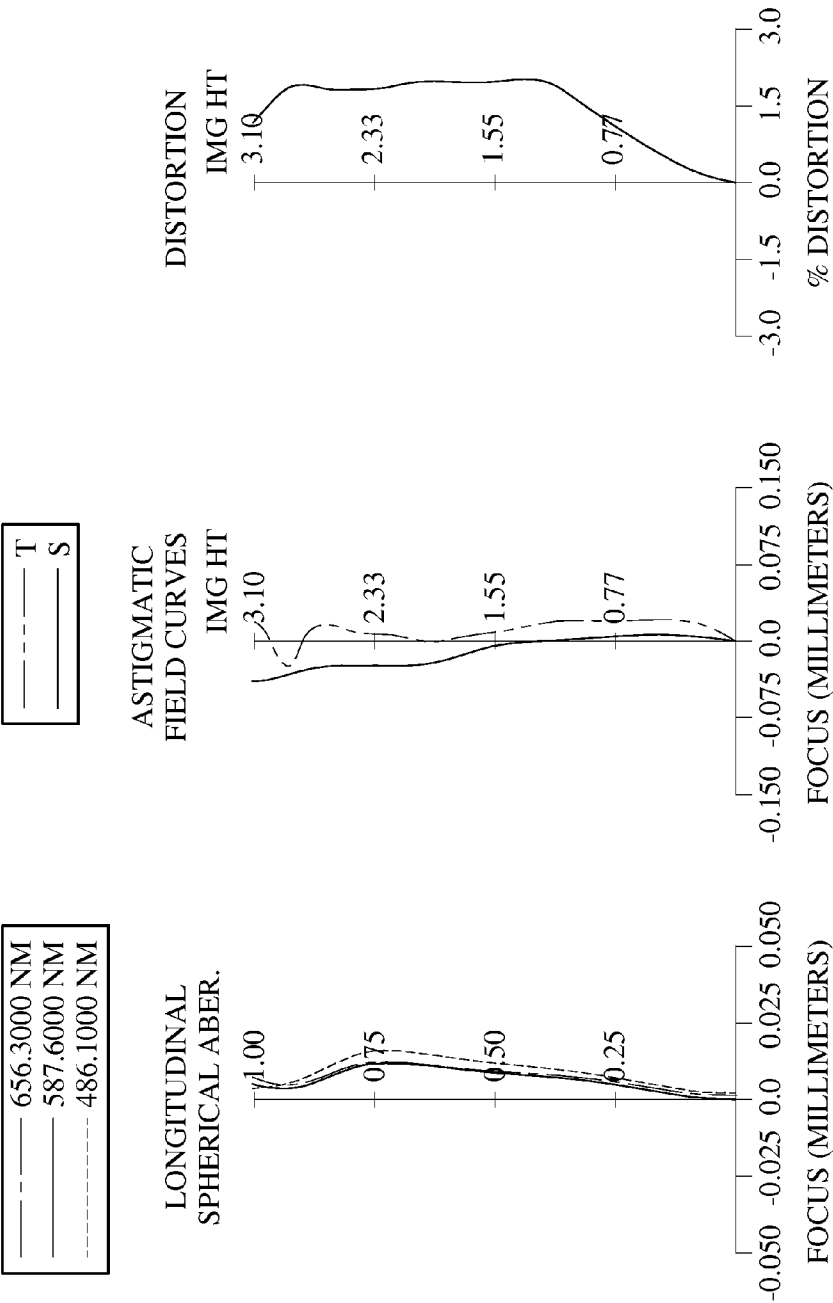


FIG. 10

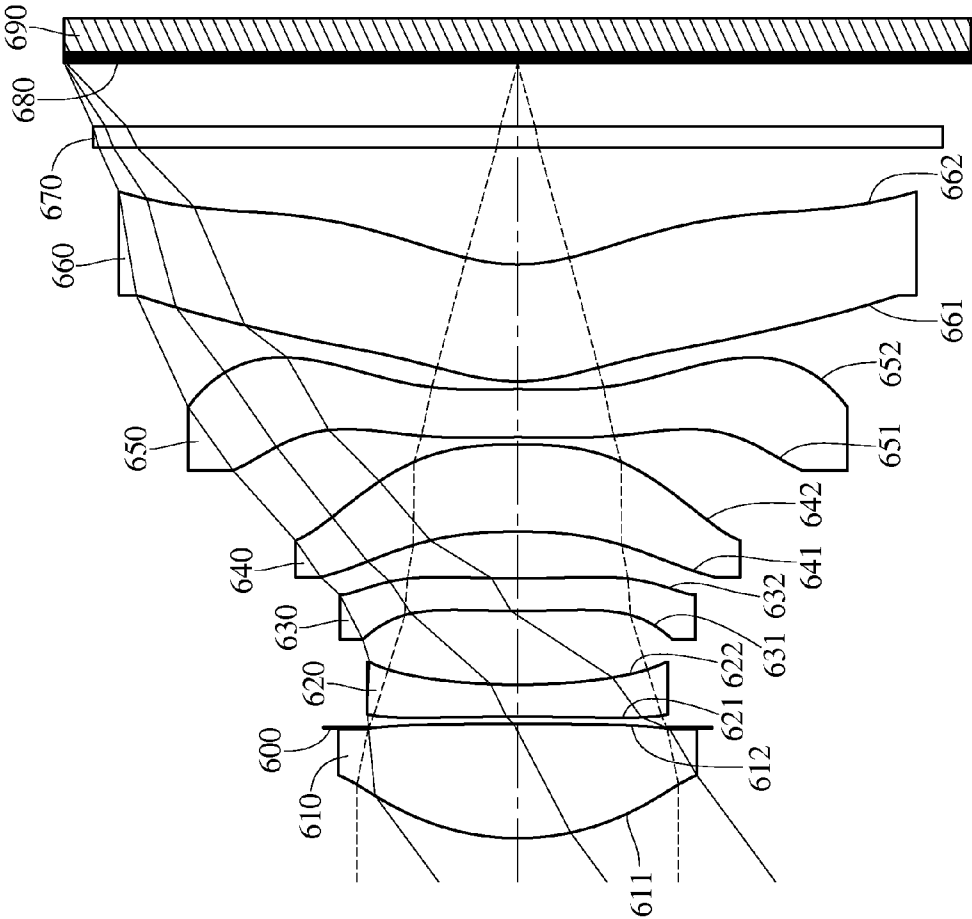


FIG. 11

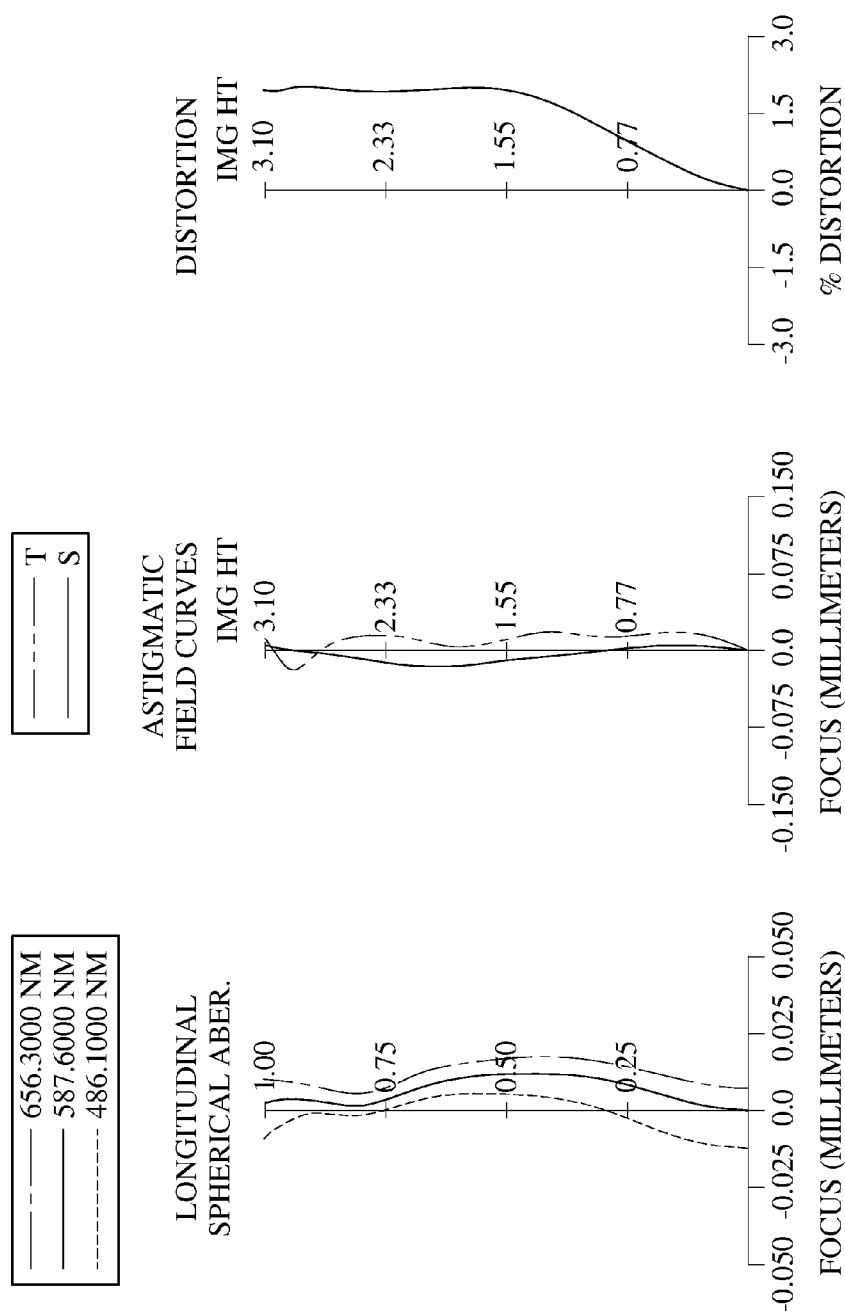


FIG. 12

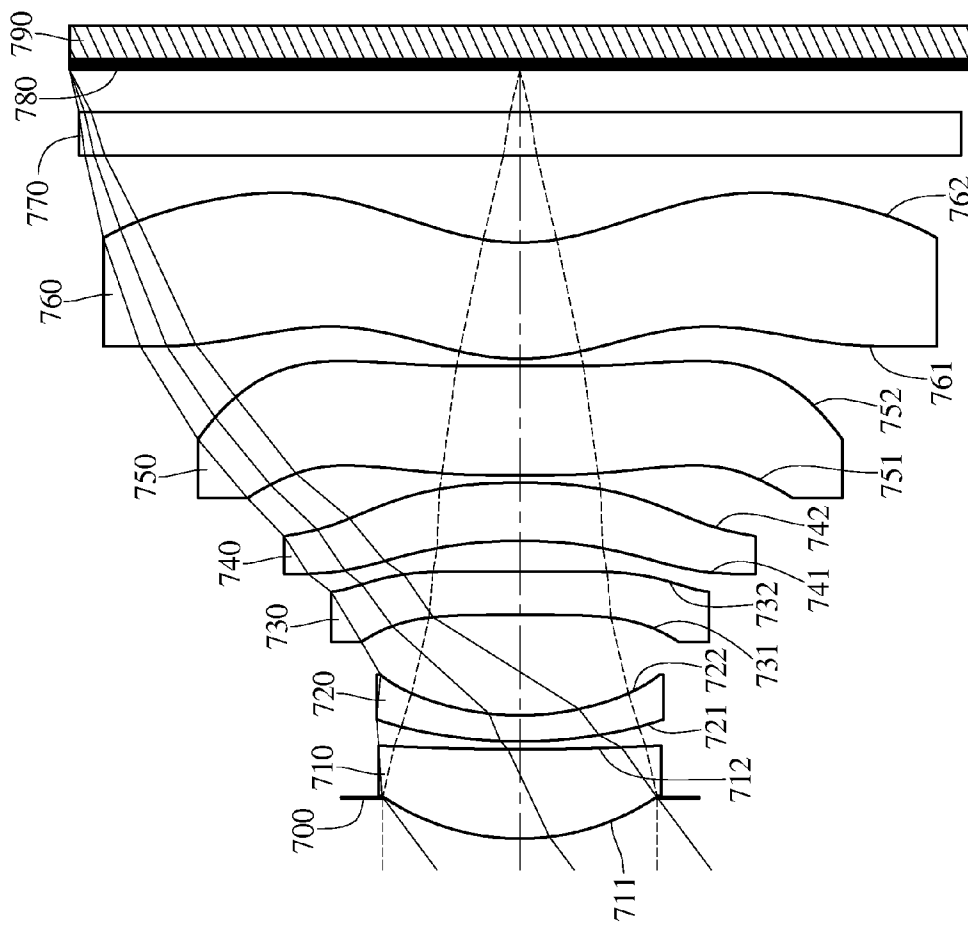
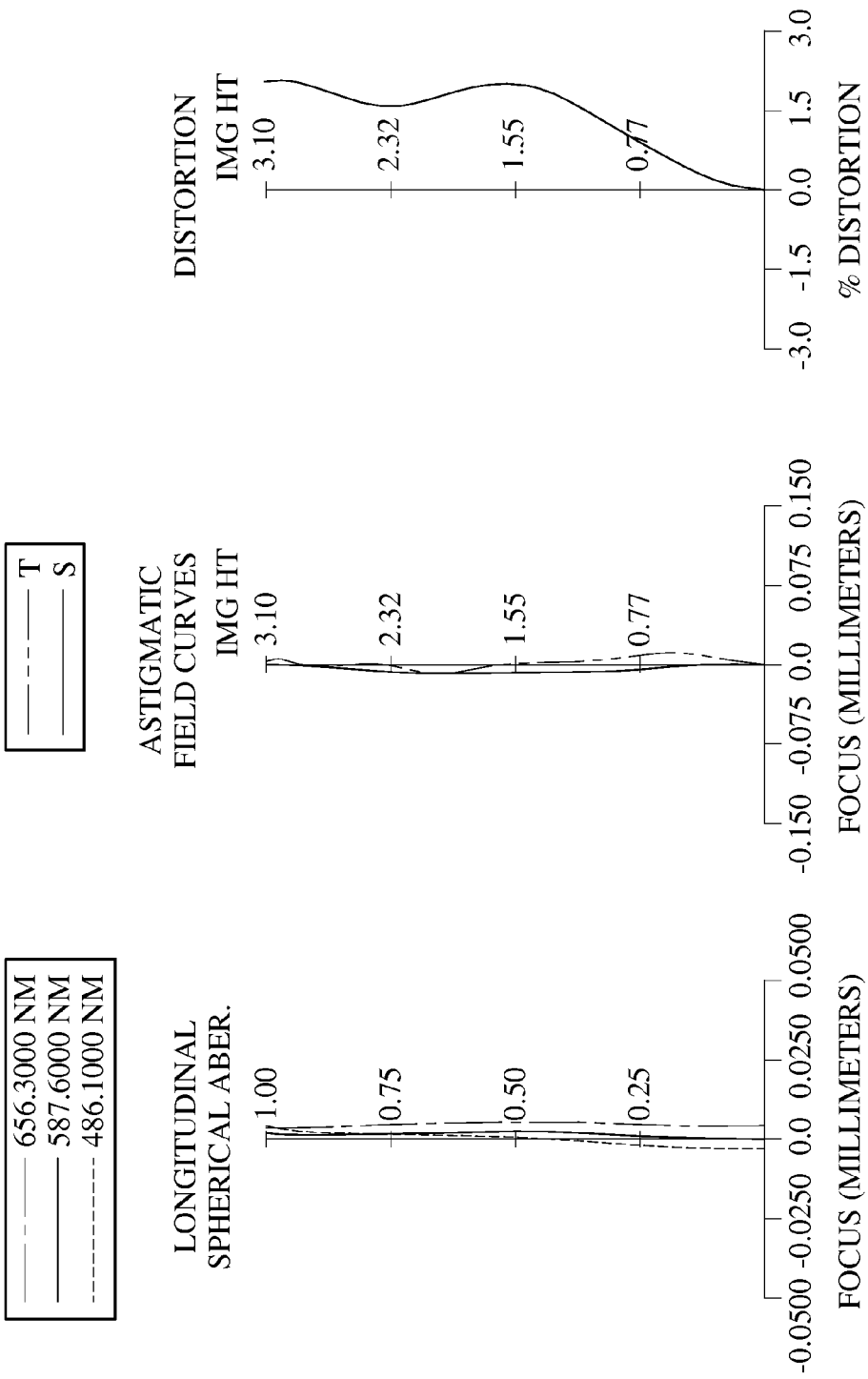


FIG. 13



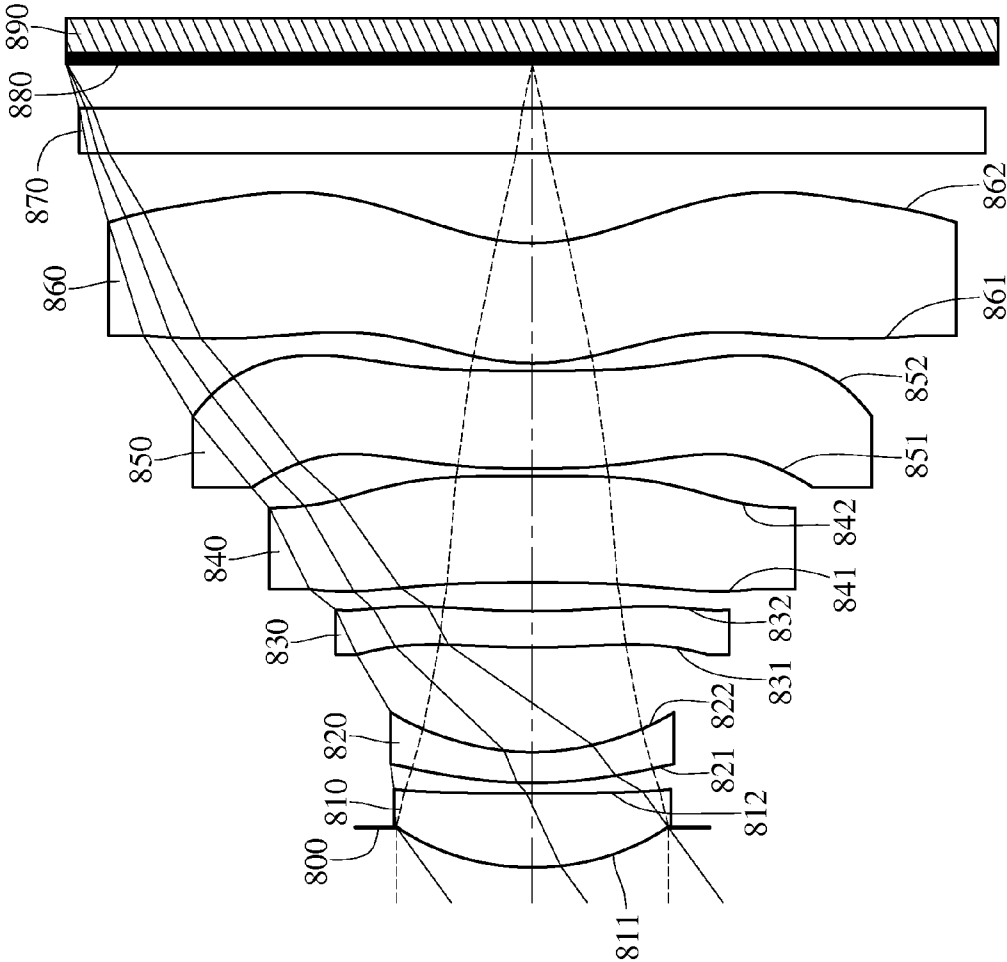


FIG. 15

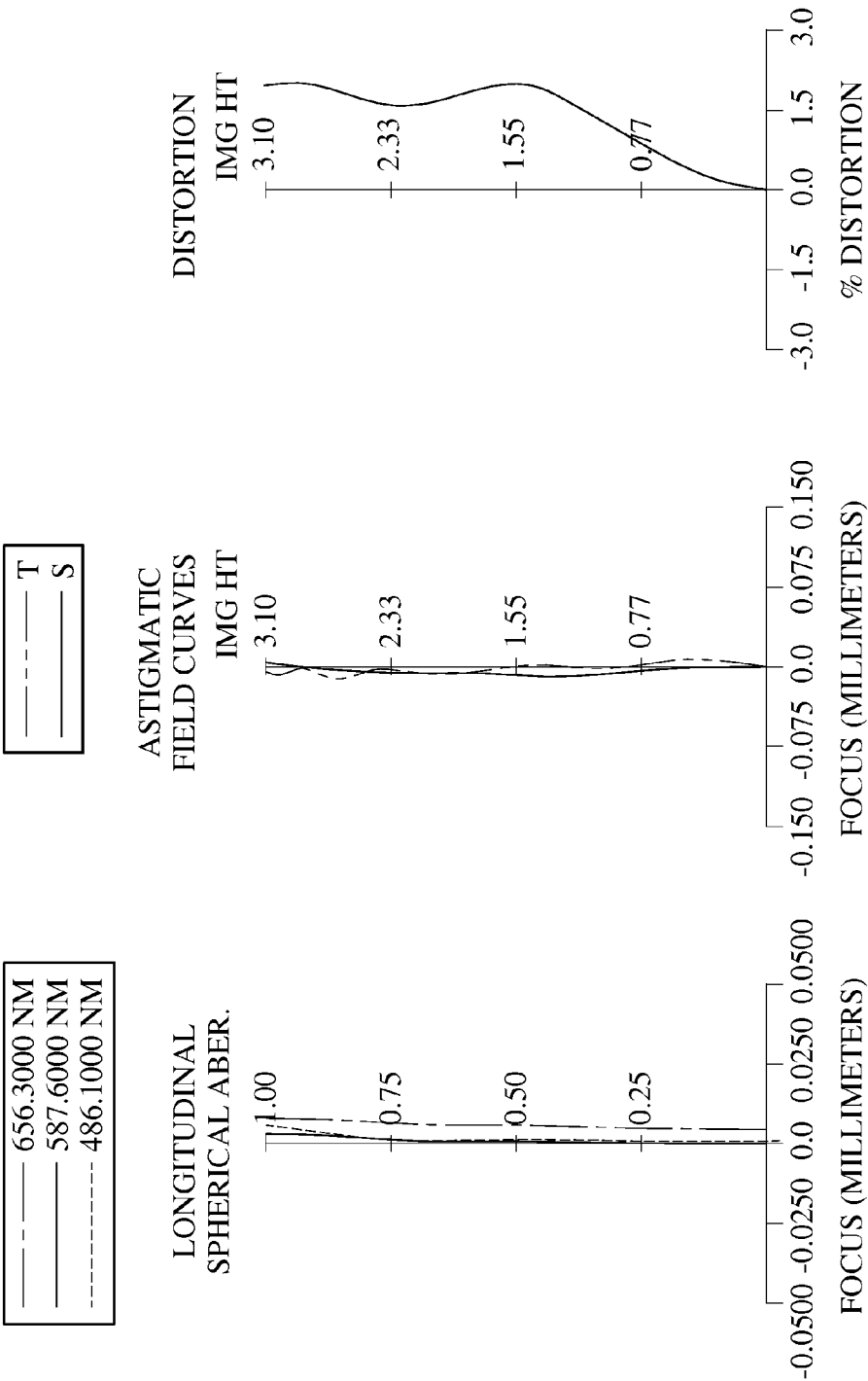


FIG. 16

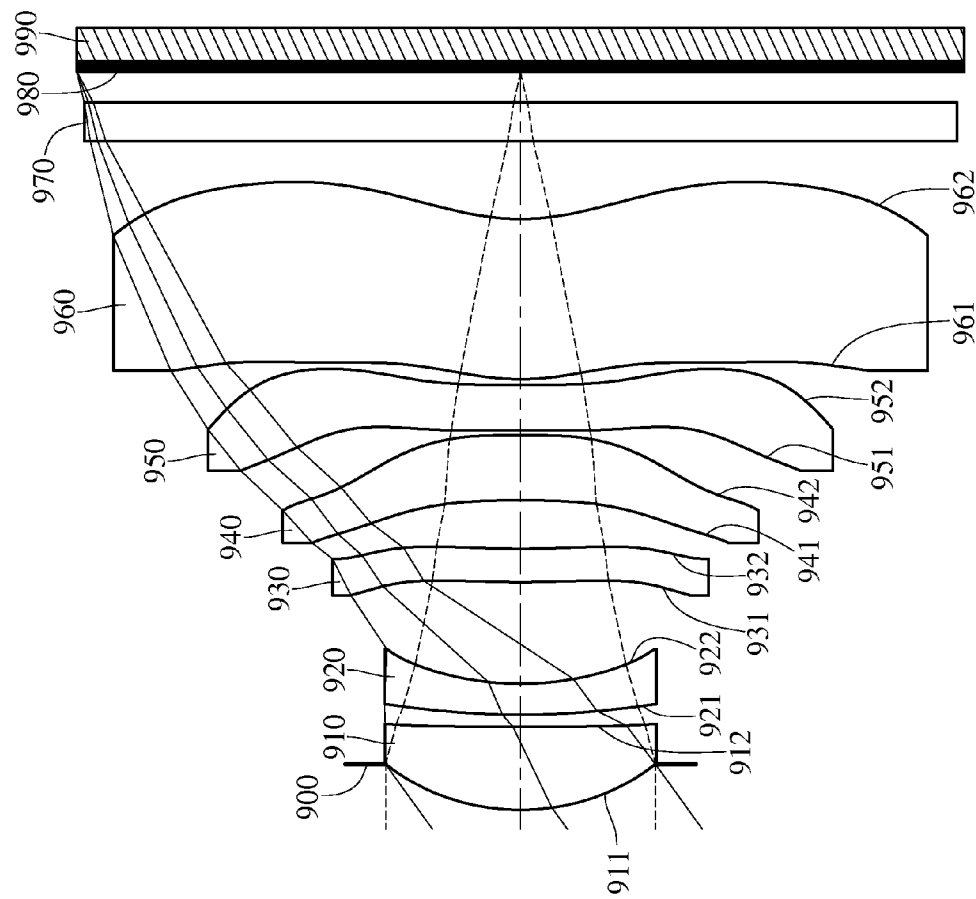


FIG. 17

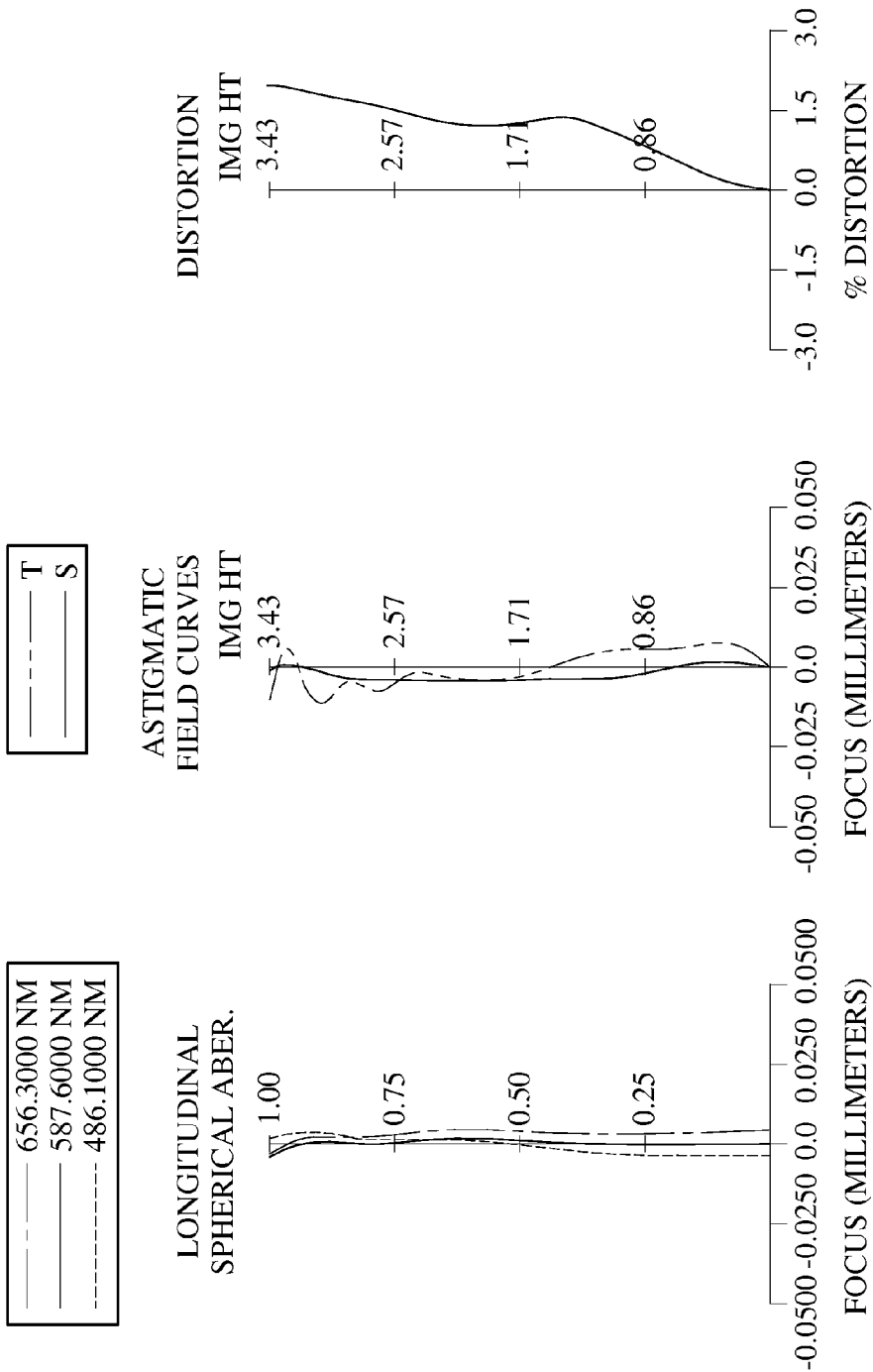


FIG. 18

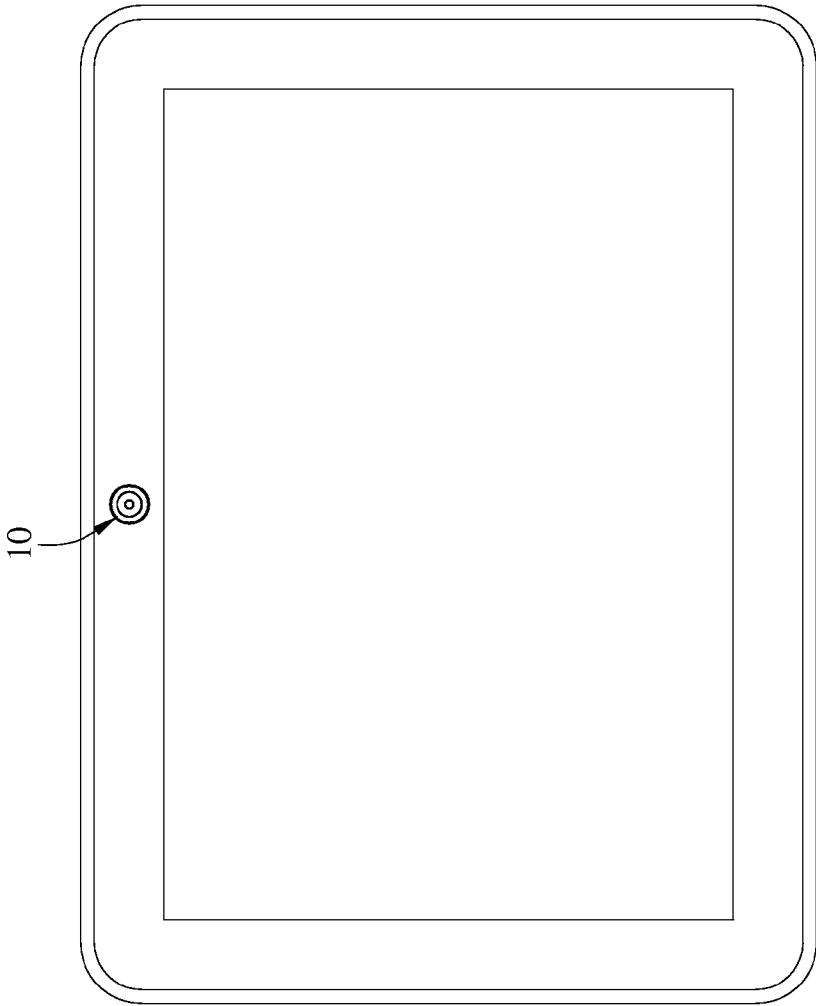


FIG. 20

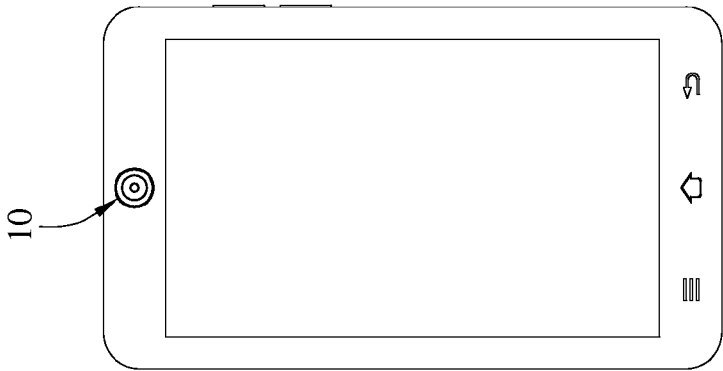


FIG. 19

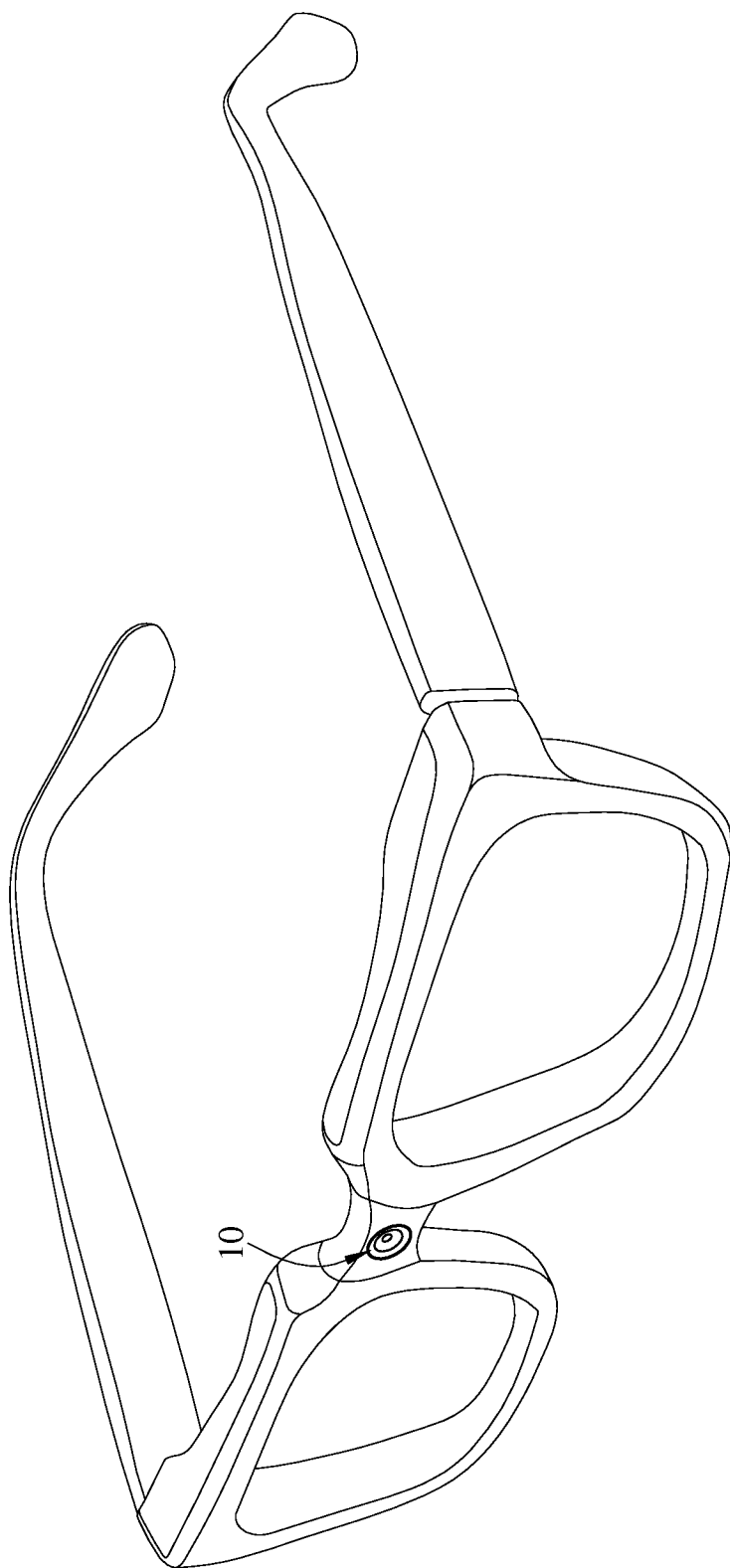


FIG. 21

PHOTOGRAPHING OPTICAL LENS ASSEMBLY, IMAGE CAPTURING UNIT AND MOBILE TERMINAL

RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Application Serial Number 103123211, filed Jul. 4, 2014, which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] 1. Technical Field

[0003] The present disclosure relates to a photographing optical lens assembly, an image capturing unit and a mobile terminal, more particularly to a photographing optical lens assembly and an image capturing unit applicable to a mobile terminal.

[0004] 2. Description of Related Art

[0005] In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

[0006] A conventional optical system employed in a portable electronic product mainly adopts a five-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. However, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

[0007] Other conventional compact optical systems with six-element lens structure are developed to enhance resolution and image quality. However, the optical systems with six-element lens structure have severe aberrations and high sensitivity. Furthermore, for the optical system, the image may be overly curved to be defocused in an off-axis region. Therefore, the imaging capability and the image quality are limited.

SUMMARY

[0008] According to one aspect of the present disclosure, a photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element has negative refractive power. The fourth lens element has negative refractive power. The fifth lens element has refractive power. The sixth lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and both of the object-side surface and the

image-side surface of the sixth lens element are aspheric. The photographing optical lens assembly has a total of six lens elements with refractive power. When a focal length of the photographing optical lens assembly is f , a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , a focal length of the sixth lens element is f_6 , a curvature radius of an image-side surface of the fifth lens element is R_{10} , and the following conditions are satisfied:

$$0 < f_3 * f_4 / f_6; \text{ and}$$

$$f / R_{10} < 0.0.$$

[0009] According to another aspect of the present disclosure, an image capturing unit includes the photographing optical lens assembly according to the aforementioned aspect and an image sensor, wherein the image sensor is disposed on the image side of the photographing optical lens assembly.

[0010] According to still another aspect of the present disclosure, a mobile terminal includes the image capturing unit according to the aforementioned aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

[0012] FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure;

[0013] FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment;

[0014] FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure;

[0015] FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment;

[0016] FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure;

[0017] FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment;

[0018] FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure;

[0019] FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment;

[0020] FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure;

[0021] FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment;

[0022] FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure;

[0023] FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment;

[0024] FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure;

[0025] FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment;

[0026] FIG. 15 is a schematic view of an image capturing unit according to the 8th embodiment of the present disclosure;

[0027] FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 8th embodiment;

[0028] FIG. 17 is a schematic view of an image capturing unit according to the 9th embodiment of the present disclosure;

[0029] FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 9th embodiment;

[0030] FIG. 19 shows a mobile terminal according to one embodiment;

[0031] FIG. 20 shows a mobile terminal according to another embodiment; and

[0032] FIG. 21 shows a mobile terminal according to still another embodiment.

DETAILED DESCRIPTION

[0033] A photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The photographing optical lens assembly has a total of six lens elements with refractive power.

[0034] The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. Therefore, it is favorable for balancing the arrangement of the refractive powers so as to avoid overloading the positive refractive power on one single lens element, thereby avoiding resulting in excessive aberration. It is also favorable for reducing the sensitivity of the photographing optical lens assembly.

[0035] The second lens element has refractive power. Therefore, it is favorable for correcting the aberration from the first lens element.

[0036] The third lens element has negative refractive power. Therefore, it is favorable for further correcting the aberration of the photographing optical lens assembly.

[0037] The fourth lens element with negative refractive power can have an object-side surface being concave in a paraxial region thereof. Therefore, it is favorable for effectively correcting the astigmatism and the aberration of the photographing optical lens assembly.

[0038] The fifth lens element can have positive refractive power. The fifth lens element can have an object-side surface being convex in a paraxial region thereof, and the object-side surface of the fifth lens element can have at least one concave shape in an off-axis region thereof. Therefore, it is favorable for avoiding excessive spherical aberrations and reducing the astigmatism. Furthermore, it is also favorable for correcting the aberration of the off-axis.

[0039] The sixth lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in an off-axis region thereof. The image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof. Therefore, it is favorable for the principal

point of the photographing optical lens assembly being positioned away from the image side of the photographing optical lens assembly and reducing the total track length of the photographing optical lens assembly so as to keep a compact size thereof. Furthermore, it is favorable for effectively reducing the incident angle of the light projecting onto the image sensor, so that it is favorable for improving the image-sensing efficiency of the image sensor and further correcting the aberration of the off-axis.

[0040] When a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , a focal length of the sixth lens element is f_6 , the following condition is satisfied: $0 < f_3 \cdot f_4 / f_6$. Therefore, it is favorable for correcting image distortion, so that the image is more planar from a paraxial region thereof to an off-axis region thereof.

[0041] When a focal length of the photographing optical lens assembly is f , a curvature radius of an image-side surface of the fifth lens element is R_{10} , the following condition is satisfied: $f / R_{10} < 0.0$. Therefore, it is favorable for adjusting the curvature radius of the image-side surface of the fifth lens element so as to correct the aberration of the photographing optical lens assembly. Preferably, the following condition is satisfied: $-3.0 < f / R_{10} < 0.0$.

[0042] When a curvature radius of an object-side surface of the second lens element is R_3 , a curvature radius of an image-side surface of the second lens element is R_4 , the following condition is satisfied: $0 < R_4 / R_3$. Therefore, it is favorable for adjusting the refractive power of the second lens element so as to correct the aberration from the first lens element.

[0043] When an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , an axial distance between the fourth lens element and the fifth lens element is T_{45} , an axial distance between the fifth lens element and the sixth lens element is T_{56} , the following condition is satisfied: $0.75 < T_{23} / (T_{12} + T_{34} + T_{45} + T_{56}) < 2.50$. Therefore, it is favorable for adjusting the axial distances between any two adjacent lens elements so as to reduce a total track length of the photographing optical lens assembly. It is thereby favorable for keeping the photographing optical lens assembly compact.

[0044] When an Abbe number of the third lens element is V_3 , the following condition is satisfied: $V_3 < 25$. Therefore, it is favorable for correcting the chromatic aberration of the photographing optical lens assembly.

[0045] When the focal length of the third lens element is f_3 , the focal length of the fourth lens element is f_4 , the following condition is satisfied: $0.4 < f_3 / f_4 < 1.2$. Therefore, it is favorable for reducing the sensitivity of the photographing optical lens assembly so as to further improve the image quality.

[0046] When an Abbe number of the second lens element is V_2 , the Abbe number of the third lens element is V_3 , an Abbe number of the fourth lens element is V_4 , the following condition is satisfied: $50 < V_2 + V_3 + V_4 < 120$. Therefore, it is favorable for correcting the chromatic aberration of the photographing optical lens assembly.

[0047] When a central thickness of the fifth lens element is CT_5 , a central thickness of the sixth lens element is CT_6 , the following condition is satisfied: $0.5 < CT_5 / CT_6 < 1.0$. Therefore, it is favorable for adjusting the thicknesses of the fifth lens element and the sixth lens element so as to reduce the total track length of the photographing optical lens assembly.

[0048] When the axial distance between the third lens element and the fourth lens element is T34, the axial distance between the fourth lens element and the fifth lens element is T45, the axial distance between the fifth lens element and the sixth lens element is T56, the following condition is satisfied: $1.0 < T34/(T45+T56)$. Therefore, it is favorable for assembling the photographing optical lens assembly so as to keep the photographing optical lens assembly compact.

[0049] When the focal length of the photographing optical lens assembly is f, a curvature radius of the image-side surface of the sixth lens element is R12, the following condition is satisfied: $0.20 < R12/f < 0.50$. Therefore, it is favorable for the principal point of the photographing optical lens assembly being positioned away from the image side of the photographing optical lens assembly so as to reduce a back focal length of the photographing optical lens assembly.

[0050] According to the photographing optical lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the photographing optical lens assembly and the image surface and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the photographing optical lens assembly and thereby provides a wider field of view for the same.

[0051] When a maximal field of view of the photographing optical lens assembly is FOV, the following condition is satisfied: $73.0 [\text{deg.}] < \text{FOV} < 100.0 [\text{deg.}]$. Therefore, it is favorable for providing sufficient field of view so as to satisfy the demand of generating proper field of view to capture the image and meanwhile keeping the image undistorted.

[0052] When a focal length of the second lens element is f2, the curvature radius of the image-side surface of the second lens element is R4, the following condition is satisfied: $|f2/R4| < 5.0$. Therefore, it is favorable for properly adjusting the refractive power of the second lens element.

[0053] According to the photographing optical lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the photographing optical lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the photographing optical lens assembly can also be reduced.

[0054] According to the photographing optical lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof. Moreover, when a region of refractive power or focus of a lens element is not defined, it indicates

that the region of refractive power or focus of the lens element is in the paraxial region thereof.

[0055] According to the photographing optical lens assembly of the present disclosure, an image surface of the photographing optical lens assembly, based on the corresponding image sensor, can be flat or curved, especially a curved surface being concave facing towards the object side of the photographing optical lens assembly.

[0056] According to the photographing optical lens assembly of the present disclosure, the photographing optical lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is set for eliminating the stray light and thereby improving the image quality thereof.

[0057] According to the present disclosure, an image capturing unit is provided. The image capturing unit includes the photographing optical lens assembly according to the aforementioned photographing optical lens assembly of the present disclosure, and an image sensor, wherein the image sensor is disposed on the image side of the aforementioned photographing optical lens assembly, that is, the image sensor can be disposed on or near an image surface of the aforementioned photographing optical lens assembly. In some embodiments, the image capturing unit can further include a barrel member, a holding member or a combination thereof.

[0058] In FIG. 19, FIG. 20 and FIG. 21, an image capturing device 10 may be installed in, but not limited to, a mobile terminal, including a smart phone (FIG. 19), a tablet personal computer (FIG. 20) or a wearable device (FIG. 21). The three exemplary figures of different kinds of mobile terminal are only exemplary for showing the image capturing device of present disclosure installing in a mobile terminal and is not limited thereto. In some embodiments, the mobile terminal can further include, but not limited to, a display, a control unit, a random access memory unit (RAM), a read only memory unit (ROM) or a combination thereof.

[0059] According to the photographing optical lens assembly of the present disclosure, the photographing optical lens assembly can be optionally applied to moving focus optical systems. Furthermore, the photographing optical lens assembly is featured with good capability in aberration corrections and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals. According to the above description of the present disclosure, the following specific embodiments are provided for further explanation.

1st Embodiment

[0060] FIG. 1 is a schematic view of an image capturing unit according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 1st embodiment. In FIG. 1, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 190. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an

image surface **180**, wherein the photographing optical lens assembly has a total of six lens elements (**110-160**) with refractive power.

[0061] The first lens element **110** with positive refractive power has an object-side surface **111** being convex in a paraxial region thereof and an image-side surface **112** being convex in a paraxial region thereof. The first lens element **110** is made of plastic material and has the object-side surface **111** and the image-side surface **112** being both aspheric.

[0062] The second lens element **120** with negative refractive power has an object-side surface **121** being convex in a paraxial region thereof and an image-side surface **122** being concave in a paraxial region thereof. The second lens element **120** is made of plastic material and has the object-side surface **121** and the image-side surface **122** being both aspheric.

[0063] The third lens element **130** with negative refractive power has an object-side surface **131** being convex in a paraxial region thereof and an image-side surface **132** being concave in a paraxial region thereof. The third lens element **130** is made of plastic material and has the object-side surface **131** and the image-side surface **132** being both aspheric.

[0064] The fourth lens element **140** with negative refractive power has an object-side surface **141** being concave in a paraxial region thereof and an image-side surface **142** being convex in a paraxial region thereof. The fourth lens element **140** is made of plastic material and has the object-side surface **141** and the image-side surface **142** being both aspheric.

[0065] The fifth lens element **150** with positive refractive power has an object-side surface **151** being concave in a paraxial region thereof and an image-side surface **152** being convex in a paraxial region thereof. The fifth lens element **150** is made of plastic material and has the object-side surface **151** and the image-side surface **152** being both aspheric.

[0066] The sixth lens element **160** with positive refractive power has an object-side surface **161** being convex in a paraxial region thereof and an image-side surface **162** being concave in a paraxial region thereof. The image-side surface **162** has at least one convex shape in an off-axis region thereof. The sixth lens element **160** is made of plastic material and has the object-side surface **161** and the image-side surface **162** being both aspheric.

[0067] The IR-cut filter **170** is made of glass and located between the sixth lens element **160** and the image surface **180**, and will not affect the focal length of the photographing optical lens assembly. The image sensor **190** is disposed on or near the image surface **180** of the photographing optical lens assembly.

[0068] The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

[0069] where,

[0070] X is the relative distance between a point on the aspheric surface spaced at a distance Y from an optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

[0071] Y is the vertical distance from the point on the aspheric surface to the optical axis;

[0072] R is the curvature radius;

[0073] k is the conic coefficient; and

[0074] A_i is the i -th aspheric coefficient, and in the embodiments, i may be, but is not limited to, 4, 6, 8, 10, 12, 14 and 16.

[0075] In the photographing optical lens assembly of the image capturing unit according to the 1st embodiment, when a focal length of the photographing optical lens assembly is f , an f -number of the photographing optical lens assembly is Fno , and half of a maximal field of view of the photographing optical lens assembly is $HFOV$, these parameters have the following values: $f=4.06$ mm; $Fno=2.10$; and $HFOV=36.8$ degrees.

[0076] When an Abbe number of the third lens element **130** is $V3$, the following condition is satisfied: $V3=21.5$.

[0077] When an Abbe number of the second lens element **120** is $V2$, the Abbe number of the third lens element **130** is $V3$, an Abbe number of the fourth lens element **140** is $V4$, the following condition is satisfied: $V2+V3+V4=98.9$.

[0078] When a central thickness of the fifth lens element **150** is $CT5$, a central thickness of the sixth lens element **160** is $CT6$, the following condition is satisfied: $CT5/CT6=0.83$.

[0079] When an axial distance between the third lens element **130** and the fourth lens element **140** is $T34$, an axial distance between the fourth lens element **140** and the fifth lens element **150** is $T45$, an axial distance between the fifth lens element **150** and the sixth lens element **160** is $T56$, the following condition is satisfied: $T34/(T45+T56)=2.47$.

[0080] When an axial distance between the first lens element **110** and the second lens element **120** is $T12$, an axial distance between the second lens element **120** and the third lens element **130** is $T23$, the axial distance between the third lens element **130** and the fourth lens element **140** is $T34$, the axial distance between the fourth lens element **140** and the fifth lens element **150** is $T45$, the axial distance between the fifth lens element **150** and the sixth lens element **160** is $T56$, the following condition is satisfied: $T23/(T12+T34+T45+T56)=1.60$.

[0081] When a curvature radius of the object-side surface **121** of the second lens element **120** is $R3$, a curvature radius of the image-side surface **122** of the second lens element **120** is $R4$, the following condition is satisfied: $R4/R3=0.47$.

[0082] When a focal length of the second lens element **120** is $f2$, the curvature radius of the image-side surface **122** of the second lens element **120** is $R4$, the following condition is satisfied: $|f2/R4|=3.04$.

[0083] When a focal length of the photographing optical lens assembly is f , a curvature radius of the image-side surface **152** of the fifth lens element **150** is $R10$, the following condition is satisfied: $f/R10=-0.09$.

[0084] When a curvature radius of the image-side surface **162** of the sixth lens element **160** is $R12$, the focal length of the photographing optical lens assembly is f , the following condition is satisfied: $R12/f=0.32$.

[0085] When a focal length of the third lens element **130** is $f3$, a focal length of the fourth lens element **140** is $f4$, the following condition is satisfied: $f3/f4=1.08$.

[0086] When the focal length of the third lens element **130** is $f3$, the focal length of the fourth lens element **140** is $f4$, a focal length of the sixth lens element **160** is $f6$, the following condition is satisfied: $f3*f4/f6=347.84$.

[0087] When a maximal field of view of the photographing optical lens assembly is FOV , the following condition is satisfied: $FOV=73.6$ degrees.

[0088] The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment								
$f = 4.06$ mm, $Fno = 2.10$, $HFOV = 36.8$ deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.313					
2	Lens 1	1.604 (ASP)	0.623	Plastic	1.544	55.9	2.90	
3		-90.994 (ASP)	0.053					
4	Lens 2	3.931 (ASP)	0.220	Plastic	1.650	21.5	-5.66	
5		1.859 (ASP)	0.638					
6	Lens 3	76.158 (ASP)	0.263	Plastic	1.650	21.5	-104.22	
7		35.815 (ASP)	0.247					
8	Lens 4	-2.427 (ASP)	0.373	Plastic	1.544	55.9	-96.59	
9		-2.682 (ASP)	0.050					
10	Lens 5	-62.651 (ASP)	0.660	Plastic	1.544	55.9	337.96	
11		-46.904 (ASP)	0.050					
12	Lens 6	1.469 (ASP)	0.800	Plastic	1.535	55.7	28.94	
13		1.314 (ASP)	0.600					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.259					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

Effective radius of the object-side surface of the sixth lens element (surface 12) is 2.585 mm.

TABLE 2

Aspheric Coefficients						
	Surface #					
	2	3	4	5	6	7
k =	8.9864E-01	-1.0000E+00	-1.0000E+00	-3.1321E+00	-1.0000E+00	1.0000E+00
A4 =	-2.4137E-02	-9.2070E-03	-7.9791E-02	-1.3593E-02	-8.8211E-02	-3.6767E-02
A6 =	-2.9517E-02	4.9862E-02	1.3580E-01	8.9677E-02	-1.5335E-01	-1.6250E-01
A8 =	5.0387E-02	-1.7101E-02	-3.4163E-03	7.3806E-02	3.2134E-01	2.5970E-01
A10 =	-1.5664E-01	-1.3246E-01	-2.2636E-01	-2.1878E-01	-5.6909E-01	-3.0285E-01
A12 =	1.6846E-01	1.6661E-01	2.5327E-01	1.9017E-01	5.0951E-01	2.0852E-01
A14 =	-8.1861E-02	-7.1093E-02	-8.6633E-02	-3.6292E-02	-1.6958E-01	-6.7254E-02
A16 =	—	—	—	—	-6.7115E-03	8.0234E-03

	Surface #					
	8	9	10	11	12	13
k =	-8.7444E+00	4.5978E-01	3.0000E+00	1.8346E+00	-5.7660E+00	-3.1050E+00
A4 =	-1.0728E-01	-1.0598E-01	1.2833E-01	3.4333E-02	-7.6475E-02	-8.5138E-02
A6 =	9.9985E-02	1.4312E-01	-1.6547E-01	-2.8216E-02	-1.2132E-02	2.7223E-02
A8 =	-3.9270E-02	-1.5231E-01	1.0632E-01	-9.0163E-03	1.6196E-02	-7.4630E-03
A10 =	8.8434E-03	1.3031E-01	-4.7777E-02	1.2159E-02	-4.5786E-03	1.6209E-03
A12 =	-3.8398E-04	-5.8127E-02	1.2578E-02	-4.9504E-03	6.2229E-04	-2.3123E-04
A14 =	-1.2074E-04	1.1923E-02	-1.3610E-03	9.7256E-04	-4.2178E-05	1.7864E-05
A16 =	—	-8.6018E-04	—	-7.5608E-05	1.1379E-06	-5.5815E-07

[0089] In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

[0090] FIG. 3 is a schematic view of an image capturing unit according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 2nd embodiment. In FIG. 3, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor **290**. The photographing optical lens assembly includes, in order from an object side to an image side an aperture stop **200**, a first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens element

250, a sixth lens element **260**, an IR-cut filter **270** and an image surface **280**, wherein the photographing optical lens assembly has a total of six lens elements (**210-260**) with refractive power.

[0091] The first lens element **210** with positive refractive power has an object-side surface **211** being convex in a paraxial region thereof and an image-side surface **212** being convex in a paraxial region thereof. The first lens element **210** is made of glass material and has the object-side surface **211** and the image-side surface **212** being both aspheric.

[0092] The second lens element **220** with negative refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric.

[0093] The third lens element **230** with negative refractive power has an object-side surface **231** being convex in a paraxial region thereof and an image-side surface **232** being concave in a paraxial region thereof. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

[0094] The fourth lens element **240** with negative refractive power has an object-side surface **241** being concave in a paraxial region thereof and an image-side surface **242** being

convex in a paraxial region thereof. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric.

[0095] The fifth lens element **250** with positive refractive power has an object-side surface **251** being concave in a paraxial region thereof and an image-side surface **252** being convex in a paraxial region thereof. The fifth lens element **250** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric.

[0096] The sixth lens element **260** with positive refractive power has an object-side surface **261** being convex in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof. The image-side surface **262** of the sixth lens element **260** has at least one convex shape in an off-axis region thereof. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being both aspheric.

[0097] The IR-cut filter **270** is made of glass and located between the sixth lens element **260** and the image surface **280**, and will not affect the focal length of the photographing optical lens assembly. The image sensor **290** is disposed on or near the image surface **280** of the photographing optical lens assembly.

[0098] The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 3.86 mm, Fno = 2.05, HFOV = 38.2 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.247					
2	Lens 1	1.749 (ASP)	0.670	Glass	1.542	62.9	3.14	
3		-53.563 (ASP)	0.050					
4	Lens 2	4.255 (ASP)	0.220	Plastic	1.639	23.5	-6.68	
5		2.088 (ASP)	0.529					
6	Lens 3	6.965 (ASP)	0.234	Plastic	1.650	21.4	-41.63	
7		5.466 (ASP)	0.260					
8	Lens 4	-5.053 (ASP)	0.477	Plastic	1.544	55.9	-43.94	
9		-6.620 (ASP)	0.050					
10	Lens 5	-14.291 (ASP)	0.504	Plastic	1.544	55.9	28.66	
11		-7.550 (ASP)	0.050					
12	Lens 6	1.304 (ASP)	0.800	Plastic	1.530	55.8	18.01	
13		1.190 (ASP)	0.500					
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
15		Plano	0.473					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients						
Surface #						
	2	3	4	5	6	7
k =	7.5580E-01	-1.0000E+00	-1.0000E+00	-2.6405E+00	-1.0000E+00	1.0000E+00
A4 =	-3.0630E-02	-3.2164E-02	-6.5333E-02	-1.2474E-02	-9.3645E-02	-6.1142E-02
A6 =	1.9481E-02	3.5778E-02	1.3542E-01	8.0206E-02	-1.6681E-01	-1.8384E-01
A8 =	-1.1360E-01	3.3874E-03	-2.2410E-02	6.4539E-02	3.7163E-01	3.5694E-01
A10 =	1.4482E-01	-1.7331E-01	-2.2365E-01	-2.1905E-01	-5.7632E-01	-4.1168E-01
A12 =	-1.0789E-01	1.8059E-01	2.5345E-01	1.9098E-01	4.7025E-01	2.6565E-01
A14 =	2.2528E-02	-6.0637E-02	-7.8756E-02	-4.7777E-02	-1.7632E-01	-8.7266E-02
A16 =	—	—	—	—	1.9620E-02	1.2793E-02

TABLE 4-continued

	Aspheric Coefficients					
	Surface #					
	8	9	10	11	12	13
k =	5.1913E-01	3.0000E+00	3.0000E+00	2.7581E+00	-2.1789E+00	-2.3810E+00
A4 =	-1.1385E-01	-1.8283E-01	2.9239E-01	2.1255E-01	-1.7642E-01	-1.2605E-01
A6 =	1.0791E-01	1.3917E-01	-3.9498E-01	-2.0958E-01	2.3637E-02	4.7793E-02
A8 =	-3.9088E-02	-1.4963E-01	2.8055E-01	1.1026E-01	1.6388E-02	-1.3923E-02
A10 =	8.3081E-03	1.3217E-01	-1.2515E-01	-3.9929E-02	-7.2913E-03	2.9349E-03
A12 =	-5.4589E-04	-5.7678E-02	2.9804E-02	9.3291E-03	1.2733E-03	-3.8389E-04
A14 =	-1.6071E-04	1.1907E-02	-2.8058E-03	-1.2328E-03	-1.0683E-04	2.6875E-05
A16 =	—	-9.6409E-04	—	6.8737E-05	3.5595E-06	-7.6675E-07

[0099] In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

[0100] Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	3.86	R4/R3	0.49
Fno	2.05	f2/R4	3.20
HFOV [deg.]	38.2	f/R10	-0.51
V3	21.4	R12/f	0.31
V2 + V3 + V4	100.8	f3/f4	0.95
CT5/CT6	0.63	f3*f4/f6	101.57
T34/(T45 + T56)	2.60	FOV [deg.]	76.4
T23/(T12 + T34 + T45 + T56)	1.29		

3rd Embodiment

[0101] FIG. 5 is a schematic view of an image capturing unit according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 3rd embodiment. In FIG. 5, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 390. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, an IR-cut filter 370 and an image surface 380, wherein the photographing optical lens assembly has a total of six lens elements (310-360) with refractive power.

[0102] The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being convex in a paraxial region thereof. The first lens element 310 is made of plastic material and has the object-side surface 311 and the image-side surface 312 being both aspheric.

[0103] The second lens element 320 with positive refractive power has an object-side surface 321 being concave in a paraxial region thereof and an image-side surface 322 being convex in a paraxial region thereof. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being both aspheric.

[0104] The third lens element 330 with negative refractive power has an object-side surface 331 being concave in a paraxial region thereof and an image-side surface 332 being convex in a paraxial region thereof. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being both aspheric.

[0105] The fourth lens element 340 with negative refractive power has an object-side surface 341 being concave in a paraxial region thereof and an image-side surface 342 being convex in a paraxial region thereof. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being both aspheric.

[0106] The fifth lens element 350 with positive refractive power has an object-side surface 351 being convex in a paraxial region thereof and an image-side surface 352 being convex in a paraxial region thereof. The object-side surface 351 has at least one concave shape in an off-axis region thereof. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being both aspheric.

[0107] The sixth lens element 360 with positive refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof. The image-side surface 362 has at least one convex shape in an off-axis region thereof. The sixth lens element 360 is made of plastic material and has the object-side surface 361 and the image-side surface 362 being both aspheric.

[0108] The IR-cut filter 370 is made of glass and located between the sixth lens element 360 and the image surface 380, and will not affect the focal length of the photographing optical lens assembly. The image sensor 390 is disposed on or near the image surface 380 of the photographing optical lens assembly.

[0109] The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment								
f = 3.78 mm, Fno = 2.45, HFOV = 38.7 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.025					
2	Lens 1	3.428 (ASP)	0.261	Plastic	1.544	55.9	5.13	
3		-14.563 (ASP)	0.109					
4	Lens 2	-22.976 (ASP)	0.648	Plastic	1.544	55.9	8.50	
5		-3.880 (ASP)	0.509					
6	Lens 3	-2.493 (ASP)	0.361	Plastic	1.650	21.5	-5.00	
7		-11.232 (ASP)	0.349					
8	Lens 4	-1.356 (ASP)	0.251	Plastic	1.650	21.5	-10.58	
9		-1.812 (ASP)	0.050					
10	Lens 5	10.759 (ASP)	0.552	Plastic	1.544	55.9	6.29	
11		-4.903 (ASP)	0.050					
12	Lens 6	1.225 (ASP)	0.800	Plastic	1.535	55.7	17.66	
13		1.087 (ASP)	0.900					
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
15		Plano	0.461					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients						
	Surface #					
	2	3	4	5	6	7
k =	-1.0575E+00	-1.0000E+00	-1.0000E+00	2.9139E+00	-1.0000E+00	1.0000E+00
A4 =	-7.3952E-02	-1.0151E-01	-7.9219E-02	-1.1500E-01	-2.4907E-01	-4.5534E-02
A6 =	9.2524E-03	2.8041E-02	6.6111E-02	-3.9186E-02	-1.3695E-01	-2.7067E-01
A8 =	-2.6365E-01	1.7865E-02	-3.0374E-02	6.7489E-02	4.4794E-01	5.8599E-01
A10 =	5.3329E-01	-1.5967E-01	-1.7240E-01	-1.9218E-01	-5.5924E-01	-6.0981E-01
A12 =	-6.2921E-01	2.7345E-01	3.3569E-01	1.8971E-01	4.5495E-01	3.2007E-01
A14 =	3.8876E-01	2.5556E-02	-1.2831E-01	-7.0734E-02	-1.8141E-01	-8.6163E-02
A16 =	—	—	—	—	2.3351E-02	1.0195E-02

	Surface #					
	8	9	10	11	12	13
k =	-7.2472E+00	-1.0138E+01	3.0000E+00	2.7581E+00	-1.8452E+00	-2.5045E+00
A4 =	-7.5398E-02	-1.4552E-01	1.9296E-01	1.4620E-01	-1.1535E-01	-8.3751E-02
A6 =	1.1097E-01	1.4735E-01	-2.3483E-01	-6.9970E-02	4.8337E-03	1.8745E-02
A8 =	-4.7094E-02	-1.5026E-01	1.4967E-01	-6.6446E-03	9.4998E-03	-1.4374E-03
A10 =	1.7156E-03	1.3212E-01	-6.4894E-02	1.6674E-02	-2.7717E-03	-2.0840E-04
A12 =	-2.9087E-03	-5.7650E-02	1.5895E-02	-7.0386E-03	3.5329E-04	6.0514E-05
A14 =	3.0996E-04	1.1881E-02	-1.5721E-03	1.3975E-03	-2.2156E-05	-5.4091E-06
A16 =	—	-1.0246E-03	—	-1.0935E-04	5.5447E-07	1.6898E-07

[0110] In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

[0111] Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	3.78	R4/R3	0.17
Fno	2.45	f2/R4	2.19
HFOV [deg.]	38.7	f/R10	-0.77
V3	21.5	R12/f	0.29
V2 + V3 + V4	98.9	f3/f4	0.47
CT5/CT6	0.88	f3*f4/f6	3.00
T34/(T45 + T56)	3.39	FOV [deg.]	77.4
T23/(T12 + T34 + T45 + T56)	1.08		

4th Embodiment

[0112] FIG. 7 is a schematic view of an image capturing unit according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 4th embodiment. In FIG. 7, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 490. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image surface 480, wherein the photographing optical lens assembly has a total of six lens elements (410-460) with refractive power.

[0113] The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being both aspheric.

[0114] The second lens element 420 with positive refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being convex in a paraxial region thereof. The second lens element 420 is made of plastic material and has the object-side surface 421 and the image-side surface 422 being both aspheric.

[0115] The third lens element 430 with negative refractive power has an object-side surface 431 being concave in a paraxial region thereof and an image-side surface 432 being convex in a paraxial region thereof. The third lens element

430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being both aspheric.

[0116] The fourth lens element 440 with negative refractive power has an object-side surface 441 being concave in a paraxial region thereof and an image-side surface 442 being convex in a paraxial region thereof. The fourth lens element 440 is made of plastic material and has the object-side surface 441 and the image-side surface 442 being both aspheric.

[0117] The fifth lens element 450 with positive refractive power has an object-side surface 451 being convex in a paraxial region thereof and an image-side surface 452 being convex in a paraxial region thereof. The object-side surface 451 has at least one concave shape in an off-axis region thereof. The fifth lens element 450 is made of plastic material and has the object-side surface 451 and the image-side surface 452 being both aspheric.

[0118] The sixth lens element 460 with positive refractive power has an object-side surface 461 being convex in a paraxial region thereof and an image-side surface 462 being concave in a paraxial region thereof. The image-side surface 462 has at least one convex shape in an off-axis region thereof. The sixth lens element 460 is made of plastic material and has the object-side surface 461 and the image-side surface 462 being both aspheric.

[0119] The IR-cut filter 470 is made of glass and located between the sixth lens element 460 and the image surface 480, and will not affect the focal length of the photographing optical lens assembly. The image sensor 490 is disposed on or near the image surface 480 of the photographing optical lens assembly.

[0120] The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment							
f = 3.70 mm, Fno = 2.38, HFOV = 39.3 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.030				
2	Lens 1	3.087 (ASP)	0.261	Plastic	1.544	55.9	10.64
3		6.415 (ASP)	0.109				
4	Lens 2	7.688 (ASP)	0.648	Plastic	1.544	55.9	4.46
5		-3.438 (ASP)	0.509				
6	Lens 3	-2.579 (ASP)	0.361	Plastic	1.650	21.5	-4.74
7		-16.608 (ASP)	0.349				
8	Lens 4	-1.266 (ASP)	0.251	Plastic	1.608	25.7	-8.43
9		-1.807 (ASP)	0.050				
10	Lens 5	9.866 (ASP)	0.552	Plastic	1.544	55.9	5.51
11		-4.219 (ASP)	0.050				
12	Lens 6	1.179 (ASP)	0.800	Plastic	1.535	55.7	15.39
13		1.050 (ASP)	0.900				
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
15		Plano	0.447				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients						
Surface #						
	2	3	4	5	6	7
k =	-2.3664E+00	-1.3891E+01	-1.0000E+00	2.0176E+00	-1.0000E+00	-1.0000E+00
A4 =	-8.4448E-02	-1.4954E-01	-1.2079E-01	-1.1734E-01	-2.5123E-01	-2.7076E-02
A6 =	9.6698E-02	5.1470E-03	-1.0423E-03	-4.2873E-02	-1.5015E-01	-2.6266E-01
A8 =	-5.8481E-01	-3.8746E-02	5.7560E-03	9.3931E-02	4.5875E-01	5.7127E-01
A10 =	1.2846E+00	-2.6886E-02	-1.7873E-01	-1.9448E-01	-5.6152E-01	-5.9551E-01
A12 =	-1.5774E+00	7.7080E-02	3.2189E-01	1.7135E-01	4.5001E-01	3.2062E-01
A14 =	8.2614E-01	8.4268E-02	-1.3286E-01	-6.0246E-02	-1.9106E-01	-9.0182E-02
A16 =	—	—	—	—	3.3663E-02	1.1189E-02

Surface #						
	8	9	10	11	12	13
k =	-6.1262E+00	-6.7051E+00	3.0000E+00	2.7581E+00	-1.8645E+00	-2.6846E+00
A4 =	-4.4943E-02	-1.5243E-01	2.1919E-01	2.3709E-01	-1.2220E-01	-7.4698E-02
A6 =	1.1271E-01	1.4513E-01	-2.2769E-01	-1.1842E-01	6.5914E-03	1.1434E-02
A8 =	-5.2082E-02	-1.5100E-01	1.2617E-01	-1.3689E-03	1.0046E-02	2.6160E-03
A10 =	3.5582E-04	1.3173E-01	-5.1404E-02	2.0173E-02	-3.1938E-03	-1.3286E-03
A12 =	-1.8934E-03	-5.7782E-02	1.1922E-02	-8.6350E-03	4.4254E-04	2.1631E-04
A14 =	1.6823E-03	1.1877E-02	-1.1176E-03	1.6648E-03	-3.0192E-05	-1.6124E-05
A16 =	—	-9.1711E-04	—	-1.2530E-04	8.1357E-07	4.6028E-07

[0121] In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

[0122] Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	3.70	R4/R3	-0.45
Fno	2.38	f2/R4	1.30
HFOV [deg.]	39.3	f/R10	-0.88
V3	21.5	R12/f	0.28
V2 + V3 + V4	103.1	f3/f4	0.56
CT5/CT6	0.69	f3*f4/f6	2.60
T34/(T45 + T56)	3.49	FOV [deg.]	78.6
T23/(T12 + T34 + T45 + T56)	0.91		

5th Embodiment

[0123] FIG. 9 is a schematic view of an image capturing unit according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 5th embodiment. In FIG. 9, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 590. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element 510, an aperture stop 500, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 570 and an

image surface 580, wherein the photographing optical lens assembly has a total of six lens elements (510-560) with refractive power.

[0124] The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof. The first lens element 510 is made of plastic material and has the object-side surface 511 and the image-side surface 512 being both aspheric.

[0125] The second lens element 520 with positive refractive power has an object-side surface 521 being concave in a paraxial region thereof and an image-side surface 522 being convex in a paraxial region thereof. The second lens element 520 is made of plastic material and has the object-side surface 521 and the image-side surface 522 being both aspheric.

[0126] The third lens element 530 with negative refractive power has an object-side surface 531 being concave in a paraxial region thereof and an image-side surface 532 being convex in a paraxial region thereof. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being both aspheric.

[0127] The fourth lens element 540 with negative refractive power has an object-side surface 541 being concave in a paraxial region thereof and an image-side surface 542 being convex in a paraxial region thereof. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being both aspheric.

[0128] The fifth lens element 550 with positive refractive power has an object-side surface 551 being convex in a paraxial region thereof and an image-side surface 552 being convex in a paraxial region thereof. The object-side surface 551 has at least one concave shape in an off-axis region thereof. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being both aspheric.

[0129] The sixth lens element 560 with positive refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being

concave in a paraxial region thereof. The image-side surface **562** has at least one convex shape in an off-axis region thereof. The sixth lens element **560** is made of plastic material and has the object-side surface **561** and the image-side surface **562** being both aspheric.

[0130] The IR-cut filter **570** is made of glass and located between the sixth lens element **560** and the image surface

580, and will not affect the focal length of the photographing optical lens assembly. The image sensor **590** is disposed on or near the image surface **580** of the photographing optical lens assembly.

[0131] The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 3.09 mm, Fno = 2.28, HFOV = 44.7 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	1.571 (ASP)	0.343	Plastic	1.544	55.9	4.38	
2		4.255 (ASP)	0.050					
3	Ape. Stop	Plano	0.279					
4	Lens 2	-9.909 (ASP)	0.473	Plastic	1.544	55.9	5.00	
6		-2.172 (ASP)	0.171					
7	Lens 3	-1.936 (ASP)	0.220	Plastic	1.650	21.5	-4.09	
8		-7.441 (ASP)	0.208					
9	Lens 4	-1.803 (ASP)	0.362	Plastic	1.650	21.5	-8.05	
10		-2.969 (ASP)	0.050					
11	Lens 5	5.303 (ASP)	0.461	Plastic	1.544	55.9	4.42	
12		-4.275 (ASP)	0.050					
13	Lens 6	1.090 (ASP)	0.608	Plastic	1.570	40.0	76.91	
14		0.891 (ASP)	0.700					
15	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
16		Plano	0.365					
17	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	1.9747E+00	-1.2329E+01	-1.9680E+01	1.8897E+00	1.0255E+00	2.6887E+00
A4 =	-5.7799E-02	2.0526E-02	-1.0603E-01	-2.5022E-01	-4.8772E-01	-1.6217E-01
A6 =	-4.1505E-02	-2.7997E-01	-2.6523E-01	-3.3393E-03	1.0365E-01	1.0208E-01
A8 =	-4.9925E-02	1.3698E+00	5.3946E-01	-4.8525E-01	-1.0014E+00	-1.9009E-01
A10 =	-4.4309E-01	-4.7914E+00	-1.7609E+00	1.5621E+00	4.8207E+00	1.1741E-01
A12 =	9.3519E-01	7.4926E+00	1.8695E+00	-1.8049E+00	-6.6950E+00	2.1678E-01
A14 =	-9.9108E-01	-4.6809E+00	-1.0420E+00	5.6977E-01	3.5094E+00	-3.1124E-01
A16 =	—	—	—	-3.7768E-02	-6.7761E-01	1.1021E-01

Surface #						
	8	9	10	11	12	13
k =	-2.0000E+01	1.4663E+00	-4.3365E-01	3.0000E+00	-2.1050E+00	-3.0175E+00
A4 =	-2.7674E-01	-2.5445E-01	1.6178E-01	3.0638E-01	-2.3856E-01	-9.1749E-02
A6 =	1.0638E+00	3.9524E-01	-1.3529E-01	-1.9249E-01	1.0089E-01	-3.6585E-03
A8 =	-1.8247E+00	-1.3114E-01	5.9515E-02	5.1572E-02	-7.9669E-02	1.5865E-02
A10 =	1.3638E+00	-3.1729E-01	-3.4473E-02	-1.3452E-02	4.5980E-02	-5.7818E-03
A12 =	-4.6270E-01	3.4703E-01	1.6163E-02	5.9931E-03	-1.2903E-02	9.7675E-04
A14 =	7.0607E-02	-1.2444E-01	-3.9016E-03	-1.5049E-03	1.7114E-03	-8.1564E-05
A16 =	-6.3605E-03	1.4705E-02	3.5821E-04	1.3162E-04	-8.7307E-05	2.6984E-06

[0132] In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

[0133] Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment			
f [mm]	3.09	R4/R3	0.22
Fno	2.28	f2/R4	2.30
HFOV [deg.]	44.7	f/R10	-0.72
V3	21.5	R12/f	0.29
V2 + V3 + V4	98.9	f3/f4	0.51
CT5/CT6	0.76	f3*f4/f6	0.43
T34/(T45 + T56)	2.08	FOV [deg.]	89.4
T23/(T12 + T34 + T45 + T56)	0.27		

6th Embodiment

[0134] FIG. 11 is a schematic view of an image capturing unit according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 6th embodiment. In FIG. 11, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 690. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element 610, an aperture stop 600, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image surface 680, wherein the photographing optical lens assembly has a total of six lens elements (610-660) with refractive power.

[0135] The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The first lens element

610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

[0136] The second lens element 620 with negative refractive power has an object-side surface 621 being concave in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric.

[0137] The third lens element 630 with negative refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

[0138] The fourth lens element 640 with negative refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being convex in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric.

[0139] The fifth lens element 650 with positive refractive power has an object-side surface 651 being concave in a paraxial region thereof and an image-side surface 652 being convex in a paraxial region thereof. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric.

[0140] The sixth lens element 660 with positive refractive power has an object-side surface 661 being convex in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof. The image-side surface 662 has at least one convex shape in an off-axis region thereof. The sixth lens element 660 is made of plastic material and has the object-side surface 661 and the image-side surface 662 being both aspheric.

[0141] The IR-cut filter 670 is made of glass and located between the sixth lens element 660 and the image surface 680, and will not affect the focal length of the photographing optical lens assembly. The image sensor 690 is disposed on or near the image surface 680 of the photographing optical lens assembly.

[0142] The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment								
f = 4.11 mm, Fno = 1.87, HFOV = 36.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	1.741 (ASP)	0.780	Plastic	1.544	55.9	3.24	
2		118.873 (ASP)	-0.025					
3	Ape. Stop	Plano	0.075					
4	Lens 2	-67.735 (ASP)	0.220	Plastic	1.639	23.5	-6.58	
5		4.489 (ASP)	0.504					
6	Lens 3	8.052 (ASP)	0.221	Plastic	1.639	23.5	-75.15	
7		6.822 (ASP)	0.322					
8	Lens 4	-3.802 (ASP)	0.595	Plastic	1.544	55.9	-44.67	
9		-4.756 (ASP)	0.050					
10	Lens 5	-9.697 (ASP)	0.329	Plastic	1.544	55.9	1454.94	
11		-9.694 (ASP)	0.052					
12	Lens 6	1.263 (ASP)	0.800	Plastic	1.535	55.7	10.53	
13		1.268 (ASP)	0.800					

TABLE 11-continued

6th Embodiment f = 4.11 mm, Fno = 1.87, HFOV = 36.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
14	IR-cut filter	Plano	0.145	Glass	1.517	64.2	—
15		Plano	0.430				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients						
	Surface #					
	1	2	4	5	6	7
k =	6.4753E-01	-1.0000E+00	-1.0000E+00	-2.0000E+01	-1.0000E+00	1.0000E+00
A4 =	-2.0290E-02	-7.6502E-02	-8.2842E-02	-4.5686E-03	-1.4581E-01	-1.1697E-01
A6 =	1.2197E-02	1.4277E-01	2.5162E-01	1.6486E-01	-9.3509E-02	-7.0881E-02
A8 =	-5.3723E-02	-1.6745E-01	-3.2066E-01	-3.2605E-01	1.1090E-01	1.7357E-01
A10 =	5.1906E-02	9.3589E-02	2.6216E-01	4.4299E-01	-9.4063E-02	-2.4735E-01
A12 =	-2.9213E-02	-2.7954E-02	-1.2303E-01	-3.3240E-01	-1.1869E-01	1.8760E-01
A14 =	2.5100E-03	3.8796E-03	3.1066E-02	1.0985E-01	2.0298E-01	-6.1716E-02
A16 =	—	—	—	—	-7.0123E-02	7.7293E-03

	Surface #					
	8	9	10	11	12	13
k =	-3.3526E+00	2.9007E+00	-2.0000E+01	-2.0000E+01	-2.7986E+00	-3.1723E+00
A4 =	-1.6507E-01	-4.5874E-01	2.4877E-01	3.3473E-01	-2.2592E-01	-1.1429E-01
A6 =	1.3288E-01	6.3344E-01	-2.3280E-01	-2.8926E-01	1.4878E-01	5.1415E-02
A8 =	-4.4940E-02	-6.8376E-01	1.0355E-01	1.3684E-01	-5.3186E-02	-1.6342E-02
A10 =	9.7447E-03	5.0727E-01	-2.9150E-02	-4.0892E-02	1.1371E-02	3.5716E-03
A12 =	-1.4420E-03	-2.2874E-01	4.7852E-03	7.4416E-03	-1.4486E-03	-4.7945E-04
A14 =	1.1259E-04	5.6589E-02	-3.3060E-04	-7.4944E-04	1.0184E-04	3.5181E-05
A16 =	—	-5.8554E-03	—	3.1845E-05	-3.0538E-06	-1.0791E-06

[0143] In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

[0144] Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	4.11	R4/R3	-0.07
Fno	1.87	f2/R4	1.47
HFOV [deg.]	36.5	f/R10	-0.42
V3	23.5	R12/f	0.31
V2 + V3 + V4	102.9	f3/f4	1.68
CT5/CT6	0.41	f3*f4/f6	318.80
T34/(T45 + T56)	3.16	FOV [deg.]	73.0
T23/(T12 + T34 + T45 + T56)	1.06		

7th Embodiment

[0145] FIG. 13 is a schematic view of an image capturing unit according to the 7th embodiment of the present disclosure.

FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 7th embodiment. In FIG. 13, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 790. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 770 and an image surface 780, wherein the photographing optical lens assembly has a total of six lens elements (710-760) with refractive power.

[0146] The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof. The first lens element 710 is made of plastic material and has the object-side surface 711 and the image-side surface 712 being both aspheric.

[0147] The second lens element 720 with negative refractive power has an object-side surface 721 being convex in a paraxial region thereof and an image-side surface 722 being concave in a paraxial region thereof. The second lens element

720 is made of plastic material and has the object-side surface **721** and the image-side surface **722** being both aspheric.

[0148] The third lens element **730** with negative refractive power has an object-side surface **731** being concave in a paraxial region thereof and an image-side surface **732** being concave in a paraxial region thereof. The third lens element **730** is made of plastic material and has the object-side surface **731** and the image-side surface **732** being both aspheric.

[0149] The fourth lens element **740** with negative refractive power has an object-side surface **741** being concave in a paraxial region thereof and an image-side surface **742** being convex in a paraxial region thereof. The fourth lens element **740** is made of plastic material and has the object-side surface **741** and the image-side surface **742** being both aspheric.

[0150] The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex in a paraxial region thereof and an image-side surface **752** being convex in a paraxial region thereof. The object-side surface **751** has at least one concave shape in an off-axis region

thereof. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being both aspheric.

[0151] The sixth lens element **760** with positive refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof. The image-side surface **762** has at least one convex shape in an off-axis region thereof. The sixth lens element **760** is made of plastic material and has the object-side surface **761** and the image-side surface **762** being both aspheric.

[0152] The IR-cut filter **770** is made of glass and located between the sixth lens element **760** and the image surface **780**, and will not affect the focal length of the photographing optical lens assembly. The image sensor **790** is disposed on or near the image surface **780** of the photographing optical lens assembly.

[0153] The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment							
f = 4.06 mm, Fno = 2.15, HFOV = 36.8 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.284				
2	Lens 1	1.690 (ASP)	0.616	Plastic	1.570	57.0	3.20
3		20.184 (ASP)	0.053				
4	Lens 2	3.124 (ASP)	0.180	Plastic	1.660	20.0	-7.38
5		1.860 (ASP)	0.692				
6	Lens 3	-63.505 (ASP)	0.299	Plastic	1.660	20.0	-30.09
7		28.945 (ASP)	0.211				
8	Lens 4	-3.533 (ASP)	0.401	Plastic	1.570	57.0	-44.14
9		-4.280 (ASP)	0.050				
10	Lens 5	19.499 (ASP)	0.752	Plastic	1.570	57.0	28.09
11		-88.302 (ASP)	0.050				
12	Lens 6	1.557 (ASP)	0.800	Plastic	1.570	57.0	21.48
13		1.451 (ASP)	0.600				
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
15		Plano	0.289				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients						
Surface #						
	2	3	4	5	6	7
k =	8.9393E-01	-1.0000E+00	-1.0000E+00	-2.8890E+00	-2.0000E+01	-1.0000E+00
A4 =	-1.8268E-02	5.7053E-03	-5.1438E-02	-6.5785E-04	-8.4019E-02	-4.9547E-02
A6 =	-1.9705E-02	1.0203E-02	8.0476E-02	6.8381E-02	-9.7969E-02	-1.1128E-01
A8 =	2.2302E-02	-1.9756E-04	-1.2866E-02	2.3556E-02	1.7443E-01	1.7026E-01
A10 =	-6.2155E-02	-5.4348E-02	-8.6134E-02	-8.1402E-02	-2.4446E-01	-1.4271E-01
A12 =	5.7357E-02	5.9614E-02	1.0167E-01	7.3324E-02	1.7211E-01	5.9724E-02
A14 =	-2.7041E-02	-2.4544E-02	-3.4813E-02	-6.9998E-03	-4.7130E-02	-4.4581E-03
A16 =	—	—	—	2.9534E-03	-1.9302E-03	—
Surface #						
	8	9	10	11	12	13
k =	-8.8724E-01	2.1241E+00	-2.0000E+01	-2.0000E+01	-3.5627E+00	-1.3510E+00
A4 =	-7.1687E-02	-1.0798E-01	1.1496E-01	7.5613E-02	-7.1326E-02	-1.2444E-01
A6 =	7.1146E-02	9.3176E-02	-1.3373E-01	-5.4860E-02	-5.3845E-03	3.6582E-02

TABLE 14-continued

Aspheric Coefficients						
A8 =	-1.7992E-02	-7.8220E-02	7.4071E-02	1.4959E-02	8.3280E-03	-9.5893E-03
A10 =	3.3787E-03	5.5664E-02	-2.7691E-02	-2.4684E-03	-1.9624E-03	1.9037E-03
A12 =	-5.0036E-04	-2.0247E-02	5.8036E-03	1.9158E-04	2.1666E-04	-2.3369E-04
A14 =	-1.0062E-04	3.4519E-03	-4.9026E-04	3.9092E-06	-1.1666E-05	1.5347E-05
A16 =	—	-2.3267E-04	—	-1.2026E-06	2.3936E-07	-4.1236E-07

[0154] In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

[0155] Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment			
f [mm]	4.06	R4/R3	0.60
Fno	2.15	f2/R4	3.97
HFOV [deg.]	36.8	f/R10	-0.05
V3	20.0	R12/f	0.36
V2 + V3 + V4	97.0	f3/f4	0.68
CT5/CT6	0.94	f3*f4/f6	61.83
T34/(T45 + T56)	2.11	FOV [deg.]	73.6
T23/(T12 + T34 + T45 + T56)	1.90		

8th Embodiment

[0156] FIG. 15 is a schematic view of an image capturing unit according to the 8th embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 8th embodiment. In FIG. 15, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 890. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 800, a first lens element 810, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, an IR-cut filter 870 and an image surface 880, wherein the photographing optical lens assembly has a total of six lens elements (810-860) with refractive power.

[0157] The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being convex in a paraxial region thereof. The first lens element 810

is made of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric.

[0158] The second lens element 820 with negative refractive power has an object-side surface 821 being convex in a paraxial region thereof and an image-side surface 822 being concave in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric.

[0159] The third lens element 830 with negative refractive power has an object-side surface 831 being convex in a paraxial region thereof and an image-side surface 832 being concave in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric.

[0160] The fourth lens element 840 with negative refractive power has an object-side surface 841 being concave in a paraxial region thereof and an image-side surface 842 being concave in a paraxial region thereof. The fourth lens element 840 is made of plastic material and has the object-side surface 841 and the image-side surface 842 being both aspheric.

[0161] The fifth lens element 850 with positive refractive power has an object-side surface 851 being convex in a paraxial region thereof and an image-side surface 852 being convex in a paraxial region thereof. The object-side surface 851 has at least one concave shape in an off-axis region thereof. The fifth lens element 850 is made of plastic material and has the object-side surface 851 and the image-side surface 852 being both aspheric.

[0162] The sixth lens element 860 with positive refractive power has an object-side surface 861 being convex in a paraxial region thereof and an image-side surface 862 being concave in a paraxial region thereof. The image-side surface 862 has at least one convex shape in an off-axis region thereof. The sixth lens element 860 is made of plastic material and has the object-side surface 861 and the image-side surface 862 being both aspheric.

[0163] The IR-cut filter 870 is made of glass and located between the sixth lens element 860 and the image surface 880, and will not affect the focal length of the photographing optical lens assembly. The image sensor 890 is disposed on or near the image surface 880 of the photographing optical lens assembly.

[0164] The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment						
f = 4.17 mm, Fno = 2.30, HFOV = 36.1 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Abbe # Focal Length
0	Object	Plano	Infinity			
1	Ape. Stop	Plano	-0.267			

TABLE 15-continued

8th Embodiment f = 4.17 mm, Fno = 2.30, HFOV = 36.1 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
2	Lens 1	1.712 (ASP)	0.496	Plastic	1.544	55.9	3.13	
3		-246.053 (ASP)	0.068					
4	Lens 2	2.903 (ASP)	0.204	Plastic	1.650	21.4	-5.99	
5		1.616 (ASP)	0.699					
6	Lens 3	5.944 (ASP)	0.240	Plastic	1.650	21.4	-30.85	
7		4.512 (ASP)	0.192					
8	Lens 4	-24.923 (ASP)	0.709	Plastic	1.544	55.9	-32.45	
9		61.157 (ASP)	0.050					
10	Lens 5	8.388 (ASP)	0.653	Plastic	1.544	55.9	13.91	
11		-75.485 (ASP)	0.050					
12	Lens 6	1.549 (ASP)	0.800	Plastic	1.544	55.9	46.73	
13		1.350 (ASP)	0.600					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.292					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients						
	Surface #					
	2	3	4	5	6	7
k =	1.2503E+00	-1.0000E+00	-1.0000E+00	-5.0251E+00	3.0000E+00	3.6041E-01
A4 =	-2.1281E-02	4.1822E-02	-6.8932E-02	3.1371E-02	-8.2476E-02	-8.3135E-02
A6 =	-1.0878E-02	-1.2965E-02	7.8671E-02	5.0480E-02	-8.4993E-02	-8.2339E-02
A8 =	-1.1853E-02	1.1195E-02	-5.2568E-03	-2.6538E-02	2.0326E-01	1.9641E-01
A10 =	3.4426E-03	7.0524E-03	-1.3399E-01	7.0257E-03	-2.6185E-01	-2.1790E-01
A12 =	6.2896E-04	-3.0961E-02	1.5318E-01	-1.0637E-02	1.6860E-01	1.3022E-01
A14 =	-8.3279E-03	2.1335E-02	-5.9873E-02	1.6153E-02	-3.6456E-02	-3.4861E-02
A16 =	—	—	—	—	-2.7439E-03	2.8853E-03

	Surface #					
	8	9	10	11	12	13
k =	3.0000E+00	3.0000E+00	-2.0000E+01	3.0000E+00	-2.8971E+00	-1.0146E+00
A4 =	-7.1427E-02	-1.4172E-01	1.0431E-01	1.2070E-01	-1.0092E-01	-1.7150E-01
A6 =	5.9617E-02	9.9061E-02	-1.2143E-01	-7.3599E-02	7.1030E-04	5.8674E-02
A8 =	-1.8152E-02	-8.2257E-02	6.2994E-02	1.7777E-02	1.2384E-02	-1.8256E-02
A10 =	3.5383E-03	5.5920E-02	-2.3861E-02	-2.2038E-03	-3.8178E-03	4.0365E-03
A12 =	-4.0028E-04	-2.0092E-02	5.3731E-03	1.2897E-04	5.1838E-04	-5.2936E-04
A14 =	-4.6094E-06	3.4864E-03	-4.9317E-04	-5.6370E-06	-3.3852E-05	3.6512E-05
A16 =	—	-2.3463E-04	—	3.6916E-07	8.5773E-07	-1.0219E-06

[0165] In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

[0166] Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	4.17	R4/R3	0.56
Fno	2.30	f2/R4	3.71

-continued

8th Embodiment			
HFOV [deg.]	36.1	f/R10	-0.06
V3	21.4	R12/f	0.32
V2 + V3 + V4	98.7	f3/f4	0.95
CT5/CT6	0.82	f3*f4/f6	21.42
T34/(T45 + T56)	1.92	FOV [deg.]	72.2
T23/(T12 + T34 + T45 + T56)	1.94		

9th Embodiment

[0167] FIG. 17 is a schematic view of an image capturing unit according to the 9th embodiment of the present disclosure. FIG. 18 shows, in order from left to right, spherical

aberration curves, astigmatic field curves and a distortion curve of the image capturing unit according to the 9th embodiment. In FIG. 17, the image capturing unit includes the photographing optical lens assembly (its reference numeral is omitted) of the present disclosure and an image sensor 990. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 900, a first lens element 910, a second lens element 920, a third lens element 930, a fourth lens element 940, a fifth lens element 950, a sixth lens element 960, an IR-cut filter 970 and an image surface 980, wherein the photographing optical lens assembly has a total of six lens elements (910-960) with refractive power.

[0168] The first lens element 910 with positive refractive power has an object-side surface 911 being convex in a paraxial region thereof and an image-side surface 912 being concave in a paraxial region thereof. The first lens element 910 is made of plastic material and has the object-side surface 911 and the image-side surface 912 being both aspheric.

[0169] The second lens element 920 with negative refractive power has an object-side surface 921 being convex in a paraxial region thereof and an image-side surface 922 being concave in a paraxial region thereof. The second lens element 920 is made of plastic material and has the object-side surface 921 and the image-side surface 922 being both aspheric.

[0170] The third lens element 930 with negative refractive power has an object-side surface 931 being convex in a paraxial region thereof and an image-side surface 932 being concave in a paraxial region thereof. The third lens element 930 is made of plastic material and has the object-side surface 931 and the image-side surface 932 being both aspheric.

[0171] The fourth lens element 940 with negative refractive power has an object-side surface 941 being concave in a paraxial region thereof and an image-side surface 942 being convex in a paraxial region thereof. The fourth lens element 940 is made of plastic material and has the object-side surface 941 and the image-side surface 942 being both aspheric.

[0172] The fifth lens element 950 with negative refractive power has an object-side surface 951 being concave in a paraxial region thereof and an image-side surface 952 being convex in a paraxial region thereof. Both of the object-side surface 951 and the image-side surface 952 have at least one inflection point. The fifth lens element 950 is made of plastic material and has the object-side surface 951 and the image-side surface 952 being both aspheric.

[0173] The sixth lens element 960 with positive refractive power has an object-side surface 961 being convex in a paraxial region thereof and an image-side surface 962 being concave in a paraxial region thereof. The image-side surface 962 has at least one convex shape in an off-axis region thereof. The sixth lens element 960 is made of plastic material and has the object-side surface 961 and the image-side surface 962 being both aspheric.

[0174] The IR-cut filter 970 is made of glass and located between the sixth lens element 960 and the image surface 980, and will not affect the focal length of the photographing optical lens assembly. The image sensor 990 is disposed on or near the image surface 980 of the photographing optical lens assembly.

[0175] The detailed optical data of the 8th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

TABLE 17

9th Embodiment								
f = 4.65 mm, Fno = 2.23, HFOV = 35.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.351					
2	Lens 1	1.758 (ASP)	0.642	Plastic	1.544	55.9	3.30	
3		78.128 (ASP)	0.092					
4	Lens 2	5.186 (ASP)	0.240	Plastic	1.650	21.5	-6.11	
5		2.209 (ASP)	0.786					
6	Lens 3	7.805 (ASP)	0.258	Plastic	1.650	21.5	-132.74	
7		7.065 (ASP)	0.374					
8	Lens 4	-10.356 (ASP)	0.503	Plastic	1.544	55.9	-32.72	
9		-25.176 (ASP)	0.040					
10	Lens 5	-13.109 (ASP)	0.352	Plastic	1.544	55.9	-58.13	
11		-22.599 (ASP)	0.040					
12	Lens 6	1.812 (ASP)	1.239	Plastic	1.535	55.7	14.97	
13		1.784 (ASP)	0.600					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.237					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 18

Aspheric Coefficients						
Surface #						
	2	3	4	5	6	7
k =	5.5820E-01	-1.0000E+00	-5.0601E+00	-5.8896E+00	3.0000E+00	-6.1937E+01
A4 =	-7.6163E-03	-6.2070E-04	-5.7505E-02	1.3038E-02	-7.1032E-02	-4.2683E-02

TABLE 18-continued

Aspheric Coefficients					
A6 =	-3.0508E-04	4.0563E-02	9.5049E-02	7.2119E-02	-3.4853E-02
A8 =	-8.4381E-03	-5.3275E-02	-8.8713E-02	-8.2578E-02	4.1022E-02
A10 =	1.0013E-02	3.5826E-02	5.0870E-02	9.5037E-02	-1.9148E-02
A12 =	-6.2738E-03	-1.0030E-03	-1.7958E-02	-5.4161E-02	-1.4833E-02
A14 =	1.1113E-03	-1.3319E-02	5.3502E-03	7.9470E-03	2.0969E-02
A16 =	7.4757E-05	5.0330E-03	-2.8352E-03	6.0702E-03	-5.8652E-03

Surface #					
	8	9	10	11	12
k =	2.9728E+00	-2.0000E+01	2.8064E+00	-4.5964E-01	-2.2440E+00
A4 =	-1.2998E-01	-3.6244E-01	1.3678E-01	2.0080E-01	-2.0169E-01
A6 =	6.0568E-02	3.6706E-01	-8.8403E-02	-1.5224E-01	8.8776E-02
A8 =	-1.0060E-02	-2.8969E-01	6.6513E-03	5.1654E-02	-1.9452E-02
A10 =	2.5186E-03	1.5777E-01	7.2116E-03	-9.0705E-03	2.1922E-03
A12 =	-4.9514E-04	-4.9719E-02	-2.3068E-03	6.8661E-04	-1.1071E-04
A14 =	-2.1167E-04	8.1587E-03	2.7883E-04	3.5484E-07	6.5573E-07
A16 =	4.5630E-05	-5.4746E-04	-1.2696E-05	-1.8736E-06	9.1100E-08

[0176] In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

[0177] Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

9th Embodiment		
f [mm]	4.65	R4/R3
Fno	2.23	f2/R4
HFOV [deg.]	35.9	f/R10
V3	21.5	R12/f
V2 + V3 + V4	98.9	f3/f4
CT5/CT6	0.28	f3*f4/f6
T34/(T45 + T56)	4.68	FOV [deg.]
T23/(T12 + T34 + T45 + T56)	1.44	

[0178] The foregoing image capturing unit is able to be installed in, but not limited to, a mobile terminal, including smart phones, tablet personal computers and wearable apparatus. According to the present disclosure, a photographing optical lens assembly provided in the disclosure includes, in order from an object side to an image side, a first lens element having positive refractive power, a second lens element having refractive power, a third lens element having negative refractive power, a fourth lens element having negative refractive power, a fifth lens element having refractive power and a sixth lens element having positive refractive power. The photographing optical lens assembly has a total of six lens elements with refractive power. When specific conditions are satisfied, it is favorable for balancing the arrangement of the refractive powers of the photographing optical lens assembly so as to reduce the aberration and the sensitivity of the photographing optical lens assembly. In particular, both of the third lens element and the fourth lens element have negative refractive power, so that it is favorable for correcting image distortion so that the image becomes more planar from a paraxial region thereof to an off-axis region thereof.

[0179] The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-18 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A photographing optical lens assembly comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element having refractive power;

a third lens element having negative refractive power;

a fourth lens element having negative refractive power;

a fifth lens element having refractive power; and

a sixth lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and both of the object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the photographing optical lens assembly has a total of six lens elements with refractive power, a focal length of the photographing optical lens assembly is f, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, a focal length of the sixth lens element is f6, a curvature radius of an image-side surface of the fifth lens element is R10, and the following conditions are satisfied:

$$0 < f3 * f4 / f6; \text{ and}$$

$$f / R10 < 0.0.$$

2. The photographing optical lens assembly of claim 1, wherein the fourth lens element has an object-side surface being concave in a paraxial region thereof.

3. The photographing optical lens assembly of claim 2, wherein a curvature radius of an object-side surface of the second lens element is R3, a curvature radius of an image-side surface of the second lens element is R4, and the following condition is satisfied:

$$0 < R4/R3.$$

4. The photographing optical lens assembly of claim 3, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and the following condition is satisfied:

$$0.75 < T23/(T12+T34+T45+T56) < 2.50.$$

5. The photographing optical lens assembly of claim 3, wherein an Abbe number of the third lens element is V3, and the following condition is satisfied:

$$V3 < 25.$$

6. The photographing optical lens assembly of claim 3, wherein the focal length of the photographing optical lens assembly is f, the curvature radius of the image-side surface of the fifth lens element is R10, and the following condition is satisfied:

$$-3.0 < f/R10 < 0.0.$$

7. The photographing optical lens assembly of claim 1, wherein the focal length of the third lens element is f3, the focal length of the fourth lens element is f4, and the following condition is satisfied:

$$0.4 < f3/f4 < 1.2.$$

8. The photographing optical lens assembly of claim 1, wherein an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, and the following condition is satisfied:

$$50 < V2+V3+V4 < 120.$$

9. The photographing optical lens assembly of claim 1, wherein a central thickness of the fifth lens element is CT5, a central thickness of the sixth lens element is CT6, and the following condition is satisfied:

$$0.5 < CT5/CT6 < 1.0.$$

10. The photographing optical lens assembly of claim 1, wherein an axial distance between the third lens element and

the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and the following condition is satisfied:

$$1.0 < T34/(T45+T56).$$

11. The photographing optical lens assembly of claim 1, wherein the fifth lens element has an object-side surface being convex in a paraxial region thereof, and the object-side surface of the fifth lens element has at least one concave shape in an off-axis region thereof.

12. The photographing optical lens assembly of claim 1, wherein a curvature radius of the image-side surface of the sixth lens element is R12, the focal length of the photographing optical lens assembly is f, and the following condition is satisfied:

$$0.20 < R12/f < 0.50.$$

13. The photographing optical lens assembly of claim 1, wherein both of an object-side surface and an image-side surface of the second lens element are aspheric, both of an object-side surface and an image-side surface of the third lens element are aspheric, both of an object-side surface and an image-side surface of the fourth lens element are aspheric, both of an object-side surface and the image-side surface of the fifth lens element are aspheric, and all of the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element are all made of plastic material.

14. The photographing optical lens assembly of claim 1, further comprising:

a stop located between an imaged object and an object-side surface of the third lens element.

15. The photographing optical lens assembly of claim 14, wherein a maximal field of view of the photographing optical lens assembly is FOV, and the following condition is satisfied:

$$73.0[\text{deg.}] < \text{FOV} < 100.0[\text{deg.}].$$

16. The photographing optical lens assembly of claim 14, wherein a focal length of the second lens element is f2, a curvature radius of an image-side surface of the second lens element is R4, and the following condition is satisfied:

$$|f2/R4| < 5.0.$$

17. The photographing optical lens assembly of claim 14, wherein the fifth lens element has positive refractive power.

18. An image capturing unit, comprising:

the photographing optical lens assembly of claim 1; and an image sensor, wherein the image sensor is disposed on the image side of the photographing optical lens assembly.

19. A mobile terminal, comprising:
the image capturing unit of claim 18.

* * * * *